Black Mayors as Role Models: Evidence from close elections in Brazil*

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Abstract

This paper studies the effects of the election of Black candidates as mayors in Brazil. Using data on candidates' self-declaration of race from the Brazilian Superior Electoral Court, I use a regression discontinuity design to compare municipal elections in which, among the first two most-voted candidates, one self-declared as White and the other one as Black (*preto* or *pardo*, following the classification from the Brazilian Institute of Geography and Statistics). I obtain evidence that the election of black candidates in such elections raises the participation of black High School students in the National High School Examination (ENEM) in the following year by approximately 30 students, and increases by 15 percentage points the proportion of black people born in the municipality that ran for state deputy in the following election. These results do not seem to be driven by administrative changes performed by the mayor, such as larger investments in education or in social programs, nor are they driven by the support given by the mayor to his political group in subsequent elections. Therefore, I argue that a plausible mechanism explaining these results is a role model effect.

Keywords: Race; Close Elections; Brazilian Elections; Role Models; Discrimination.

JEL Codes: J15; D72; D91.

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1 Introduction

Racial inequality is pervasive in Brazil and many other countries. While over 50% of the Brazilian population self-declares being black, the racial distribution of income or power in the country is far from egalitarian. In terms of political representation, less than 30% of elected candidates to the Chamber of Deputies (the low chamber of the legislative branch) self-declares as black. When it comes to education, black students are 15 percentage points less likely to graduate from High School than white students, and the rate of analphabetism is more than twice as large for the black population compared to the white population (9.1 against 3.9%). The income distribution is also unequal: considering only individuals with an occupation, black workers earn on average 57,5% of white workers' earnings (IBGE, 2019).

Inequalities such as these are present in several other aspects of Brazilian society, and stem from a complex set of reasons — related to the country's historical formation, systemic racism, and other factors. These inequalities are a problem in themselves and may have negative and lasting consequences in economic development, for example by reducing innovation and growth[1]. In particular, one factor that may contribute to such inequalities is that of differential beliefs and aspirations. If someone's beliefs (for instance, on the returns to education or on the importance of political participation) are shaped by the examples they have around them — and especially by the examples of those that they identify with —, it would be expected that black individuals have on average lower beliefs and aspirations than whites. If this is the case, those individuals might invest less in education and political participation, creating a trap of low beliefs, low aspirations, and low investments that reinforces racial inequalities (Genicot and Ray, 2017, 2020).

In this paper, I study one of the mechanisms that is usually discussed in the literature as capable, however slightly, of reducing some of these gaps by changing beliefs and aspirations: the role model effect. Theoretically, a role model is someone who influences the objectives, motivations, and decisions of a person, by acting as a model of behavior, a representation of what is possible, or as an inspiration (Morgenroth et al., 2015). More specifically, the contact with a role model with whom a person identifies (for instance, due to a shared identity such as race) may change this person's decision on important aspects of life, such as investment in education or the decision of which career to pursue. This could help to reduce racial gaps if a role model incentivizes black individuals to increase their political participation or their investment in education, for example.

This paper presents evidence of the existence of such a role model effect in the context of the election of black candidates in Brazilian municipal elections. The main idea is that the election of a black candidate as mayor may change the belief of individuals who identify with this candidate about their own chance of succeeding on a path that is traditionally not perceived (by them or by others) as feasible to them. With this new belief, these individuals

[1]For instance, Hsieh et al. (2019) discuss how the misallocation of talent due to racial and gender discrimination may have substantially reduced the economic output of the United States between 1960 and 2010, while Cook et al. (2021) argue that systemic racism and sexism hinder innovation at every stage of this process, which has negative consequences not only for the individuals who directly suffer from these phenomena but also to the economy as a whole.

may change their behavior, for instance, by investing more in education or by deciding to start a political career.

I use data from the 2016 municipal election, the first one in which Brazil's Superior Electoral Court collected data on candidates' self-declaration of race or skin color. Candidates had to answer the question *"what is your race or color?"*, by choosing among five alternatives: branco (white), *preto* (black), *pardo* (mixed-race or "brown"), *amarelo* ("yellow", the term used to refer to people of asian descent) or *indígena* (indigenous). This question and these options are standard in Brazilian statistics. Throughout the study, I consider as black all candidates who declared their race as *preto* or *pardo*. We take an approach that views race as a constructed social category (Sen and Wasow, 2016; Rose, 2022) rather than a fixed or immutable characteristic. Hence, the self-declaration of race must be understood as a proxy for the perceived social identity of the mayor and not as a biological fact — given that it is well understood and documented that racial categories do not represent biological differences between people, and may even vary with context.

Therefore, the main objective of this study is to estimate the causal effect of the election of candidates who self-declare as black for mayor. Estimating this effect, however, is challenging due to the probable endogeneity in the victory of a black candidate in a municipal election. Indeed, it is likely that municipalities in which a black candidate won the election are very different from municipalities in which a black candidate lost — and even more different from municipalities that did not have any black candidate running. To overcome this challenge, I use a regression discontinuity design. First, I obtain a sample composed only of the municipalities whose election in 2016 was such that, among the two candidates that obtained the most votes, one self-declared as black and the other one as white — we will call these "interracial" elections throughout this paper. Then, I estimate the effect of the election of a black candidate considering only a subsample of municipalities that had an interracial election decided by a short margin (a "close" election). Under the assumption that municipalities with a close interracial election are similar regardless of the election result (victory of a black or white candidate), it is possible to estimate the causal effect of the election of a black mayor.

Using this method, I study the causal effect of the election of a black mayor in two contexts. First, I consider educational decisions of high school students who reside in the municipality in which the interracial election happened. Education is one of the main channels through which a role model effect operates, and there is evidence, for instance, that female role models change undergraduate students' decision of what discipline to major in (Porter and Serra, 2019), or that same-race teachers have long-run impacts on students (Gershenson et al., Forthcoming). Differently from these studies, I analyse if a mayor — who does not have an individual personal relationship with each student's education — may act as a role model for this student, changing his or her educational investment decisions.

Specifically, I consider as outcome the enrollment by black and white high school students in Brazil's National High School Examination (ENEM), which is a high stakes exam taken annually by millions of Brazilians and that defines university admission for several universities in the country. I find that the election of a black mayor increases ENEM enrollment of black high school students in the exam in the year after the election by approximately 30

students, while it has no effect on white students. This effect is quantitatively significant since on average there are 55 black students enrolled in high school in the municipalities of the sample.

The second context studied in this paper is political participation. In the Brazilian context, this is particularly important given the under-representation of the black population in politics (Campos and Machado, 2015). The literature also frequently considers the impact of role models on political participation or aspirations. For instance, some papers analyze the impact of women politicians on adolescent girls' aspirations (Campbell and Wolbrecht, 2006; Beaman et al., 2012) or political engagement (Arvate et al., 2021). This last paper is somewhat related to this one since it considers the context of Brazil's municipal elections. However, while that paper considers how female mayors can act as role models and restricts the analysis to the effect of those politicians on voting behavior, I consider how black mayors can act as a role model on a wider range of outcomes related to education and political participation.

To study the effect of the election of black mayors on political participation, I consider the 2018 legislative election — from the perspective of the mayor, this is a "mid-term" election since a mayor's term is four years. Specifically, I consider the election for Federal Deputy (the low chamber of the Federal Legislative branch) and for State Deputy (the legislative council of each Brazilian state). I find that the election of a black candidate as mayor of a municipality in 2016 increases the proportion of individuals born in that municipality who run for State Deputy that self declare as black. There is no effect on candidates running for Federal Deputy. The increase in the proportion of black candidates, in the case of State Deputy, is around 15 percentage points. This increase does not occur by crowding out white candidates: the election of black candidates for mayor in 2016 does not affect the number of white candidates running for State Deputy, i.e., the increase in proportion is fully explained by an increase in the number of black candidates running for State Deputy.

In principle, the two sets of results just described may be explained by several different channels, and I present evidence that helps to rule out some alternative explanations. First, considering the results related to ENEM enrollment, I show that the election of a black mayor has no effect on the municipality's expenditure on education or culture, nor on school infrastructure. This suggests that the results are not driven by educational policy changes made by the mayor once elected. There is also evidence that the number of families who are beneficiaries of a conditional cash transfer program is not affected by the election of a black mayor. For the results on political participation, I show that the effect obtained is larger for inexperienced candidates and for candidates who are not from the same party as the elected mayor. This suggests that the results are not driven by the mayor favoring his or her political group. Therefore, a more plausible explanation for the results is that the black mayor is acting as a role model.

This paper is related to different strands in the literature. First, it contributes to the literature on role models by considering a different context in which role models may be relevant. This literature shows evidence that politicians may set an example for the population (Ajzenman et al., 2022; Ajzenman, 2021), and that television (or the media in general) may also act as a role model, changing people's behavior (Riley, Forthcoming; Jensen and Oster,

2009; Chong and La Ferrara, 2009; La Ferrara et al., 2012). More closely related to this work, some strands of the literature on role models explore how individuals from specific groups may act as role models for individuals who identify with them. When it comes to gender, there is evidence that female politicians increase aspirations and political engagement of women (Beaman et al., 2012; Arvate et al., 2021), as well as change voter behavior (Beaman et al., 2009); and that the exposure to women who graduated in economics may increase the likelihood that a female student enrolls in economic classes or even majors in this field, which is traditionally male-dominated (Porter and Serra, 2019).

When we consider race, on the other hand, the literature on role models is less extensive. There is considerable evidence that the matching of students and teachers of the same race brings benefits to black students (Dee, 2005; Fairlie et al., 2014; Gershenson et al., Forthcoming), but most studies – with the notable exception of Gershenson et al. (Forthcoming) – do not attempt to separate the channels through which this effect may be operating. Similar results are obtained on health by Alsan et al. (2019), who show that black patients are more likely to go to medical appointments and undergo invasive surgical procedures when treated by black doctors. Nevertheless, to the best of my knowledge, this paper presents the first evidence of black role models in the context of politics.

In this sense, this paper is also related to the literature on race and elections. There is considerable literature on this topic for the United States, but not for Brazil — in part due to data restrictions, since the first election for which there is data on candidates' racial declaration is the one of 2016. For the United States, some papers show that elections involving black candidates are different from most elections, for example, because elections with black candidates increase voter turnout (Washington, 2006). Vogl (2014) shows that this difference in turnout happens even in close elections. Although this difference could potentially be a concern in our setting, I show that the elections in which a black candidate wins by a short margin are not significantly different from the ones in which a black candidate loses by a short margin in the case of Brazil[2].

While there is somewhat extensive literature studying how elections involving black candidates differ from other elections, fewer papers consider the effects of electing a black candidate — which is the main contribution of this paper. A few exceptions that discuss this topic are Hopkins and McCabe (2012), which show that the election of a black candidate as mayor in the US leads to policies that are indistinguishable from the ones in cities not governed by a black mayor, and Nye et al. (2015), that show that the election of a black mayor increases employment among the black population, also in the US. Another related paper is Broockman (2013), which performs a correspondence study on US legislators and finds that black politicians are more likely than white politicians to respond to requests from black citizens, even if they are not from the politician's district.

Compared to this literature, the contribution of this paper is threefold. First, it considers a different institutional setting, Brazil — in fact, this is among the first papers to present causal estimates of the effect of the election of black mayors in this country. The only other paper I am aware of that does this is Rabelo et al. (2022). The authors study the effects of

[2]And, considering specifically the matter of turnout, this is unlikely to be an issue since voting in Brazil is mandatory for citizens above 18 years old.

the election of black candidates on the composition of municipal managers and on policies implemented by the mayor[3]. Studying race and elections in Brazil is relevant given the pervasive racial inequality in the country, especially the lack of political representation of the black population. Second, differently from the papers discussed, it considers the channels behind the results obtained[4] – and, in particular, argues that they can be explained by a role model effect. Third, by using a regression discontinuity design in close interracial elections for a relatively large sample of elections, it is capable of plausibly identifying the causal effect of the election of black mayors.

Finally, this paper is related to the literature that uses regression discontinuity (RD) designs to study close elections, inaugurated by Lee (2008). Several papers consider close municipal elections in Brazil (see, for example, Brollo and Troiano (2016); Klašnja and Titiunik (2017); Colonnelli et al. (2020); Bruce et al. (2022), or the already discussed Arvate et al. (2021)). Some authors question the validity of the RD identification assumption in the context of US House of Representatives elections, by showing that the assumption of continuity of baseline variables near the cutoff does not hold in this context (Caughey and Sekhon, 2011). Nevertheless, Eggers et al. (2015) discuss, both theoretically and empirically, why such assumptions do hold in most elections (including Brazilian municipal elections). I thoroughly discuss and present evidence in favor of the validity of the RD assumptions in our setting in section 6.

The remainder of this paper is organized as follows: in the next section, I present the data and Brazil's institutional background; section 3 discusses the empirical methodology; section 4 shows the paper's main results; section 5 discusses the channels that may explain the results; section 6 presents robustness exercises, evidence for the validity of the RD assumptions and discusses the external validity of the results; finally, section 7 concludes.

2 Data

The objective of this paper is to study what happens when a candidate who self-declares as black wins a municipal election in Brazil. Municipalities are the smallest political-administrative unit in Brazil for which there are autonomous elected governments. Elections take place every 4 years to elect a mayor (the chief of the municipalities' executive branch). For municipalities with less than 200 thousand voters, elections happen in one round, with simple plurality rule. For larger municipalities, mayors are elected by a majority rule with run-off. I use data from three different sources to measure the effects of the election of black mayors: electoral data from Brazil's Superior Electoral Court; educational data from

[3]Rabelo et al. (2022)'s paper complements the analysis in this paper, since it shows that the election of black candidates as mayors in a close municipal election has no effect on policies directed towards the black population of the municipality, nor on the racial composition of municipal managers, at least on the short run. Therefore, any result obtained in this paper cannot come from policy or administrative changes made by the mayor, but from a role model effect.

[4]While Broockman (2013) discusses a potential channel that differs black politicians from white politicians, he does not discuss how this difference may cause different outcomes when a district elects a black candidate.

Brazil's National High School examination; and administrative data from a set of other sources. These data are described in what follows:

2.1 Electoral Data. Electoral data are obtained from Brazil's Superior Electoral Court, the highest body of the country's electoral justice system. I focus on the 2016 municipal election, the first one in which the Superior Electoral Court asked candidate's to declare their "race or skin color"[5]. The data obtained is on the candidate's level and includes several information about candidate's characteristics (gender, age, occupation, political party, etc), as well as information about the election (number of votes obtained by each candidate, who was elected, etc). Particularly relevant in the context of this study is the information about candidates race or skin color, which is self-declared by the candidate among five available answers: *branco* (white), *indígena* (indigenous), *amarelo* ("yellow", term used to refer to people of asian descent), *preto* (black) or *pardo* (mixed-race or "brown"). These five categories are typical in Brazilian statistics, and the last two categories will be considered, throughout this study, as "black".

With the data from the 2016 elections, I construct a data-set that includes only municipalities in which, among the two candidates with the most votes, one self-declares as white, and the other self-declares as black (as discussed, *preto* or *pardo*). This type of election will be called "interracial elections" in this paper. Due to concerns about strategic behavior, the sample excludes municipalities which had a run-off election[6].

The final dataset contains 1563 elections (which represents 27,4% of all Brazilian municipalities). The distribution of self declaration of race between winners and runner-ups is displayed on Table 1. We can see that the number of interracial elections won by white candidates or black candidates is roughly the same – specifically, out of the 1563 elections, 809 were won by a white candidate, against 754 by black candidates. The elections won by white or black candidates are also distributed widely across Brazil's territory, as demonstrated by figure 1. The figure shows that there were interracial municipal elections in 2016 in all 26 Brazilian states. States in the North and Northeast have a relatively larger proportion of interracial elections, while states in the South had the smallest incidence of this type of election. Nevertheless, the figure also shows that the spatial distribution of elections won by a black candidate is similar to the spatial distribution of elections won by a white candidate. This is important when we consider the identification assumptions of the regression discontinuity design method, which will be discussed more deeply in the next section.

Finally, I also obtained candidate-level data from the 2018 election, which was a national and state election. We use this data to verify if the election of a black candidate in the 2016 municipal election had an effect on political participation. Focusing on the legislative election, I construct a variable that equals the proportion of candidates to state and federal deputy who were born in each municipality and who self-declare as black. In section 4, we use these two variables as an outcome, to verify if the election of a black candidate increases

[5]Precisely, the first election for which there exists information on candidate's racial declaration is the 2014 national election. Therefore, 2016 is the first municipal election for which such information is available.

[6]Less than 1% of the municipalities in the sample had a run-off election. For a discussion about the strategic component of run-off elections in Brazil's municipal elections, see Fujiwara et al. (2011).

the participation of black candidates in subsequent elections.

2.2 Educational Data. Apart from testing the effect of the election of a black mayor on political participation, we are also interested in studying the effect of the election of a black mayor on educational choice. To study this effect, we use data from Brazil's National High School Examination (ENEM). ENEM is an annual exam that is taken by millions of Brazilians, most of whom are high school students. The exam is used by all federal universities – and many other public and private universities – in the country as the main (in most cases, unique) criteria to decide college admissions. It is also an alternative way of obtaining a High School diploma for students that abandoned school. Therefore, ENEM is an important, high stakes exam for Brazilian students, and represents the chance to have a superior education (either by entering university or signaling high school completion).

The variable of interest in this study is the number of high school students residing in each municipality who enroll for ENEM in each year. The choice to enroll for the exam is made strictly by each individual. Hence, we expect that the decision to take or not the test depends on the individual's belief about his chance of succeeding in a more academic path (college). If the election of a black mayor updates the individual's belief about his chance of succeeding on this path, that is traditionally seen as white, we would expect an increase in ENEM enrollment after the election of a black candidate, specially for black students – that are more likely to identify with the mayor.

Using ENEM's microdata from 2017 and 2018, I obtained the number of high school students who reside in each Brazilian municipality who enrolled for ENEM. I focus on students who were in High School, since this is the age in which students typically take this test (and is arguably an age in which the decision to take ENEM or not matters the most). For 2017, out of an universe of 6,731,341 candidates, 39.6% reside in one of the municipalities from the sample; for 2018, out of 5,513,747 candidates, 27.8% reside in one of the municipalities from the sample.

Apart from ENEM's microdata, I also use data from Brazil's School Census to obtain, for each municipality, the number of students enrolled in High School, as well as information about municipal schools' infrastructure.

2.3 Other Data. Finally, I use administrative and socio-economic data at the municipality level. Data on municipalities' gross domestic product (GDP), alphabetization rate and population are obtained from Brazil's Institute of Geography and Statistics; municipal's annual expenditure by function is obtained from the System of Accounting and Fiscal Information of the Brazilian Public Sector; and data on the number of beneficiaries of *Bolsa Família*, Brazil's main conditional cash transfer program, comes from the Ministry of Citizenship. These data will be used to test the regression discontinuity's hypothesis, as covariates in some specification or in the discussion of mechanisms in section 5.

3 Empirical Strategy

The main challenge related to estimating the causal effect of the election of black mayors resides in the probable endogeneity of the victory of a black candidate. Indeed, municipalities in which a black candidate is successful are probably systematically different from municipalities in which black candidates are not successful, and even more different from municipalities in which no candidate self-declares as black. These differences – for instance, how diverse the municipality is – are not observable in the data. Therefore, a simple comparison of outcomes between municipalities with elected black candidates and the remaining municipalities would most certainly yield biased estimates for this causal effect.

To deal with this problem, I use a regression discontinuity (RD) design, comparing close interracial elections. Intuitively, the idea behind this methodology is that municipalities in which a black candidate won the election by a short margin and municipalities in which a black candidate was the runner-up by a short margin are very similar, so that the difference in the election result is quasi-random. In other words, the two groups of municipalities are a good counter-factual to one another.

Formally, following Cattaneo et al. (2019), we can think about this setting using a potential outcomes approach. Let $Y_i(0)$ and $Y_i(1)$ be the potential outcomes of municipality i when a black mayor loses or wins the election, respectively[7]. Let M_i be the margin of votes of the black candidate in the 2016 election, defined as the difference between the black candidate's vote share and the white candidate's vote share (hence, $M_i \geq 0$ means that the election winner was the candidate who self declares as black). Treatment status (a black mayor wining the election) is denoted by the dummy $B_i = \mathbb{1}\{M_i \geq 0\}$, which equals 1 when municipality i has elected a black candidate. Therefore, the observed outcome for each municipality is $Y_i = Y_i(1) \cdot B_i + Y_i(0) \cdot (1 - B_i)$.

Under the key identifying assumption that the functions $\mathbb{E}[Y_i(1)|M_i = m]$ and $\mathbb{E}[Y_i(0)|M_i = m]$ are continuous at the threshold $m = 0$, we can identify the RD treatment effect, given by $\mathbb{E}[Y_i(1) - Y_i(0)|M_i = 0]$. This is the effect of electing a black mayor for municipalities which are exactly at the threshold $M_i = 0$. Intuitively, the continuity assumption just stated means that municipalities that had interracial elections decided by a short margin (i.e., which are close to the threshold) are comparable or statistically identical – specifically, we expect that, for those close elections, the result of the election is independent of pre-determined characteristics of the municipality. Another implication of this assumption is that candidates cannot be able to perfectly manipulate the "side" of the threshold they will be at. This is expected, since municipal elections in Brazil are generally extremely competitive. Nevertheless, I present extensive evidence on the validity of the RD assumptions in section 6.

To estimate the RD effect, as is standard in the literature, I use a local linear regression, restricting the sample of municipalities with an interracial municipal election to municipali-

[7] We are restricting the analysis to the municipalities that had an interracial municipal election in 2016, i.e., municipalities in which, among the two candidates who obtained the most votes, one self-declares as white, and the other as black.

ties with $M_i \in [-h, -h]$ and estimating the equation:

$$Y_i = \alpha + \beta B_i + \gamma M_i + \lambda X_i + u_i \tag{1}$$

where Y_i is an outcome of interest for municipality i from a period after 2016 (the election year); X_i is a matrix of pre-determined municipal characteristics, and u_i is an idiosyncratic error. We are interested in $\hat{\gamma}$, the estimate of the RD treatment effect. For the main specifications, equation 1 is estimated using a triangular kernel, and the optimal bandwidth h is computed by minimizing the Mean Square Error (MSE) of the estimator, following Calonico et al. (2014). We report bias-corrected estimates and standard errors. For comparison, in most results tables, we also report local linear regressions using the optimal bandwidth computed with the method from Imbens and Kalyanaraman (2012).

Apart from estimation using a local linear regression, we also estimate a polynomial regression of degree p using the full dataset. The reason to also fit a polynomial regression is to account for the possibility that the apparent discontinuity in the data actually stems from non-lineariaties in the data. The polynomial model is represented by the equation:

$$Y_i = \sum_{k=0}^{p} \rho_k \cdot M_i^k + B_i \cdot \sum_{k=0}^{p} \varphi_k \cdot M_i^k + \delta X_i + \varepsilon_i \tag{2}$$

In the case of equation 2, the parameter of interest is φ_0, which represents the effect of the election of a black mayor. Finally, for comparison, we also present standard Ordinary Least Squares (OLS) estimates using the full sample for all results presented in the next sections.

4 Main Results

This section presents the main results from the paper. We start by describing some features of the data and discussing the balance between municipalities in which a black candidate won or lost the election – respectively, treated and control municipalities. We then present the effect of the election of a black mayor on ENEM enrollment. Finally, we present the effects of the election of a black mayor on political participation.

4.1 Descriptive Statistics. Table 2 presents some descriptive statistics of the sample, diving municipalities between the ones in which a black candidate won the 2016 municipal election and the ones in which a a black candidate was the runner-up.

The first panel of Table 2 shows information on the elected mayor's characteristics. There are some statistically significant differences in the sample between black and white mayors in terms of gender and age: black candidates who win the election are on average more likely to be male and younger. On the other hand, there is no significant difference in terms of political affiliation – mayors seem to be equally likely to be from a right or left wing party, and to be from two of Brazil's largest political parties (the Workers' Party and the Party of the Brazilian Social Democracy) – or in terms of education.

The second panel shows municipality's characteristics for years before the 2016 election. There are no significant differences between the two groups of municipalities in terms of location (as can be visually seen in figure 1), GDP, Alphabetization rate, population, number of high school students and per capita expenditure in education and culture.

The fact that there are some differences (even if only a few) between municipalities that elected a black candidate and municipalities in which a black candidate was the runner-up is not a threat to the RD continuity assumption because Table 2 considers the full sample of municipalities that had an interracial municipal election in 2016. The continuity assumption from the RD design only requires the treated and control municipalities are similar in terms of the baseline variables around the threshold (i.e., for municipalities that had a close interracial election). One way of providing evidence in favor of this hypothesis is to verify if there are discontinuities in pre-treatment variables for observations around the RD threshold. I do this in Figures A.1 through A.4 in the Appendix, which show RD estimates and 90% confidence intervals for the effect of the election of a black mayor on all pre-treatment variables. The figure shows that we cannot reject the null hypothesis that treated and control municipalities are equal across for most pre-treatment variables. In only one case (complete High School), we would reject this hypothesis, but this is probably due to chance[8]. The complete results are also on Table A.2 on the Appendix, which also shows that there are no discontinuities on lagged outcome variables.

The figure also reports 90% confidence intervals for the RD estimates, which allows us to consider an equivalence test approach as suggested by Hartman (2021). We see that the confidence interval for most pre-treatment variables is not too wide, indicating that we could reject a null hypothesis of "equivalence" which stated that the discontinuity in pre-treatment variables is large enough to interfere with the validity of the RD design. The potential discontinuities in pre-treatment variables are also shown graphically on figures A.1 through A.4 on the Appendix, which show RD plots for all pre-treatment variables just discussed. The figures reinforce the evidence just discussed on the lack of significant discontinuities in pre-treatment variable around the RD cutoff. Therefore, we have evidence in favor of the RD continuity assumption. We discuss more about the validity of the RD assumptions on section 6.

4.2 ENEM enrollment and investment in education. The first set of results reported is related to the educational choices of black students after the election of a black candidate as mayor in 2016. I use data on enrollment to Brazil's National High School Examination (ENEM) for 2017 and 2018. ENEM is an annual exam organized by Brazil's Federal Ministry of Education and taken by millions of Brazilians, most of whom are high school students. The exam is used by all federal universities (and several other public and private universities) as an admission criterion. It can also be used by high school dropouts to obtain a diploma for this level of education. Therefore, ENEM is a high stakes exam that represents the chance to have better education (either by entering university or by completing high school).

[8]When we correct the p-values for multiple testing using the approach from Benjamini and Hochberg (1995), we are unable to reject this null hypothesis.

Importantly, the decision to take this exam is generally made by each student personally. A student will choose to take the exam if the expected benefits of doing so outweigh the costs; thus, if the election of a black candidate leads to changes in the beliefs of young students who identify with this candidate, increasing their perceived probability of being successful in a more academic path, then we would expect an increase in ENEM enrollment after such an election.

Table 3 shows the estimates for the effect of the election of a black mayor in a close interracial election on ENEM enrollment by high school students in the following years. As in all result tables, I present six estimates using different specifications: first, using the whole sample, I present OLS estimates of equation (2) of a polynomial of degree one and a polynomial of degree four. The other four specifications are RD estimates, and differ on the method used to obtain the bandwidth: estimates from column (3), the preferred specification, uses Calonico et al. (2014)'s method to compute the optimal bandwidth, using a triangular kernel and reporting bias-corrected estimates and standard errors; the specification in column (4) is equivalent, but using half the optimal bandwidth; finally, specifications from columns (5) and (6) use Imbens and Kalyanaraman (2012)'s method to compute an optimal bandwidth (where the last specification considers half the optimal bandwidth). In all specifications, standard errors are clustered at the state level. The specifications do not include covariates, as is standard in the RD literature, but the results are robust to the inclusion of pre-treatment covariates, as shown on Appendix Table A.5.

For high school students who self-declare as black, the election of a black candidate increases ENEM enrollment both in 2017 and in 2018. For both years, the RD estimates are positive and statistically significant across all specifications. Considering our preferred specification (column (3)), in 2017 we estimate that the election of a black candidate increases in approximately 35 the number of black students who decide to take the exam. This effect is slightly smaller in 2018 (of 29 students), but the difference is not statistically significant. Given that the average number of black high school students in each municipality in the sample is of 56.7, this number is quantitatively significant as well. These results can also be seen graphically on Figure 4, which show an RD plot for ENEM enrollment in 2017 and 2018 by black high school students. On this figure, we can see the discontinuity on this outcome around the RD cutoff. The results are robust to several bandwidths, as shown in figure 6.

On the other hand, when we consider high school students who self-declare as white, Table 3 shows that the election of a black mayor has no effect on ENEM enrollment for this group of students. Indeed, the point-estimates are not statistically significant and have varying signals. This suggests, therefore, that the election of a black mayor on a close interracial election does not affect the decision of white students to take or not the exam. Considering the idea of a role model effect, this result was expected: a role model would only exert influence on individuals who identify with him or her, which would not be the case for students who identify as white.

However, there are several possible mechanisms that may explain the results obtained. For instance, it is possible that, once elected, a black candidate invests more on the municipality's black population. It is also possible that the election of a black candidate changes the racial consciousness of young students from the municipality, causing them to change

their racial identification. Both alternative explanations will be discussed on section 5.

4.3 Political Participation. The previous section discussed how the election of a black mayor may increase the investment on education by individuals who identify with this mayor, through an increase in enrollment for Brazil's National High School Examination. We now consider another dimension in which the election of a black mayor may have an impact: political participation.

Specifically, we consider the 2018 legislative election, the first election to happen after the 2016 municipal election. The 2018 election, as all elections that happen on the mid-term of mayor's mandates in Brazil, was a national and state election. In this type of election, the population votes for the national and state executive (president and governors, respectively), as well as for national and state legislative. We focus on the election for Federal Deputies (the national legislative low chamber) and for State Deputies (the state's legislative council). For each of these two positions, I obtain the number of candidates born in each municipality of the sample of municipalities that had an interracial election in 2016 by their race self-declaration. With this information, I construct variables for the proportion of candidates who were born in each municipality and who self-declare as black. This variable – for each of the two political positions – is the outcome that will be considered in this section.

Table 4 shows the estimates of the effect of the election of a black mayor on the 2016 municipal election on the proportion of candidates running for State and Federal deputy who were born in the municipality and self-declare as black. For State Deputies, there is some evidence that the election of a black candidate increases the proportion of black candidates: the point-estimates are positive across all specifications, but estimation is imprecise – when we add covariates, we gain precision, as shown in Table A.6 in the appendix. On the other hand, we do not have evidence that the election of a black candidate in a municipal election changes the proportion of candidates for Federal Deputy born in the municipality who self-declare as black: point-estimates are close to zero and the signal of the estimates depend on the bandwidth choice.

These results, however, consider all candidates running for the positions of State and Federal deputies. Nevertheless, the role model effect, if it exists, is expected to be stronger among less experienced candidates – since these candidates' decision to run or not in the election would be more susceptible to being influenced by the success of a black candidate in a previous election. Therefore, we also consider a sub-sample of the candidates for the 2018 election including only candidates who never held an elected office before[9]. This group of candidates can be interpreted as a group of less experienced candidates, whose decision to run or not on an election is more likely to influenced by the result of a previous election.

Panel A of Table 5 shows the RD estimates for the effect of the election of a black mayor on an interracial municipal election on the proportion of candidates for State Deputy who self-declare as black for this sub-sample of candidates. The results for Federal Deputy, which are not reported, confirm the previous observation that the election of a black mayor had no effect on the proportion of black candidates running for this position. On the other

[9]Specifically, we only have information about candidates holding office or not since 2008.

hand, as Panel A of Table 5 shows, the election of a black mayor in an interracial election increases the proportion of black candidates born in the municipality running for State Deputy. The estimates are statistically significant on most specifications, and robust to different bandwidths (Figure 7). Considering our preferred specification, we estimate that the election of a black candidate in the 2016 municipal election increases the proportion inexperienced of black candidates running for State Deputy by 16 percentage points. This effect is larger than the one obtained considering all candidates running for state deputy, which is an indirect evidence in favor of the role model mechanism: we expect less experienced candidates (the ones who never held an elective office before) to be more susceptible to being influenced by a role model. We discuss the mechanisms behind the results discussed so far in the next section.

5 Role Model Effect and Alternative Mechanisms

In the previous section, we saw that the election of a black mayor on a close interracial election increases the enrollment of black high school students on Brazil's National High School Examination (ENEM), and increases the proportion of black individuals from the mayor's municipality who decide to run for State Deputy in the following mid-term election, specially if we consider only inexperienced candidates. Considered more broadly, these two results can be interpreted, respectively, as an increase in the investment on education (or at least on the desire to invest on education, since taking ENEM is the first step to entering university or to completing High School); and as an increase in political participation of black individuals. What are, however, the channels behind these results? In this section, we investigate this question by considering the evidence in favor of several alternative explanations to the two phenomena described.

Specifically, one hypothesis is that the results are explained by a role model effect: the election of a black candidate as mayor changes individual's beliefs about their chance of succeeding on paths that were not considered by them as feasible, which changes the decision to run or not on an election or even to invest more on education. Another possibility is that the mayor, once elected, changes policies that are favorable to the black population of the municipality. In the context of the ENEM results, we would expect an increase in investment on education or on other social programs. In the context of the political participation results, the elected mayor could actively favor other members of his political group. We provide some evidence for each of these alternative explanations in what follows.

5.1 Education Expenditure and School Infrastructure. We start by discussing the evidence on the effect of the election of a black mayor on investment on education. In Brazil, municipalities are responsible for basic education, that is, for pre-school and elementary school[10]. This fact alone already suggests that the mayor does not have much power to affect

[10]This is defined in Brazil's Constitution, on article 30, subparagraph VI, which states that it is among the municipality's competencies to *"maintain, with the technical and financial cooperation of the Union and states, pre-school and elementary school education."*. For a detailed discussion on the federative organization

educational policy related to the municipality's high school education (which is generally a competency of the state) and which is the focus of this paper.

Nevertheless, we obtain evidence that the election of a black candidate on an interracial municipal election does not affect expenditure on education, at least on a short term. We use data on municipal's annual expenditure by function obtained from the System of Accounting and Fiscal Information of the Brazilian Public Sector. Table 6 shows the RD estimates for the effect of the election of a black mayor on educational expenditure in 2017 – the results for 2018, not reported here, are extremely similar. There is no evidence that the election of a black mayor increases the municipality's expenditure on education. If anything, there is some evidence of a decrease in this expenditure – this result, however, is not robust to different bandwidths or to the inclusion of covariates (see Table A.8).

We obtain a similar result for the effect of the election of a black mayor on the municipality's expenditure on culture: there is no evidence of an increase in this area as well. These results suggest that the election of a black mayor does not have an effect on the municipality's educational policy, at least considering the short run (the first and second year after the election). This is the period relevant to our analysis, since we only have data on ENEM enrollment for the years 2017 and 2018.

Apart from considering the effect of the election of a black mayor on educational expenditure, we also consider the effect on municipal school's infrastructure. This is relevant because it is possible that, even if a black mayor does not have more money to spend on education, his or hers administration could be more effective on the use of the resources. If this happens, we would see an improvement in municipal school's infrastructure. We consider two proxies for school infrastructure, obtained from Brazil's yearly School Census: the proportion of schools with a library or reading room and the proportion of schools with access to the internet. We focus on municipal schools, i.e., schools administered by the mayor's office[11]. The results are similar when we consider the entire population of schools in the municipality.

Table 7 shows the RD estimates for the two infrastructure outcomes. Once again, we do not obtain any evidence that the election of a black candidate increases the proportion of municipal schools with a library or reading room, nor with access to the internet. For both outcomes, we estimate a null effect – we only report the results for the 2017 School Census, but the results are the same for the 2019 School Census.

The evidence just discussed suggests that the election of a black mayor does not systematically change the municipality's education or culture expenditure, nor does it changes municipal schools' infrastructure. This is an argument against the explanation that the mayor, once elected, executes policies that lead to as increase in ENEM enrollment, as we would expect these policies to be related to education. However, the results just discussed are aggregate, and do not inform us about the distribution of resources within the municipality. Therefore, it is possible, for instance, that even if the expenditure on education and culture remains constant, a mayor chooses to direct more of these resources to some groups. This cannot be ruled out by our data. Nevertheless, the results just discussed reinforce the

of education policy in Brazil, see Abrucio (2010).

[11]Generally, municipalities in Brazil also have state and federal schools, as well as private schools.

hypothesis that a role model effect may be acting in the educational context. Indeed, we do not have evidence that municipalities that elected a black candidate as mayor in a close interracial election had a systematic improvement on education or an increase in investment on culture when compared with the municipalities in which a black candidate was a runner-up in such an election.

5.2 Social Programs. Apart from a direct change in educational policy, the election of a black mayor could have an impact on other policies – in particular, on social programs. In this section, we investigate this possibility considering one of Brazil's most recognized and successful social programs: the conditional cash transfer policy *Bolsa Família*. Although *Bolsa Família* is a Federal program, registry to the program is done at the municipality level, so it is possible that the election of a black candidate as mayor affects this program.

To test for this possibility, we use as outcome the average number of families who are beneficiaries of the program in each municipality in the years of 2017 and 2019. Table 8 documents the RD estimates of the effect of the election of a black candidate as mayor on the average number of *Bolsa Família* beneficiaries in each of these years. We have some weak evidence that the number of beneficiaries increases following the election of a black candidate. The estimates, however, are not robust to different bandwidths and lose significance once we add covariates (see Table A.10). Therefore, it is unlikely that this explains the results – moreover, it would not be clear why an increase in the number of Bolsa Família beneficiaries would mediate the increase in political participation or the increase in ENEM enrollment. Hence, this small effect does not explain the results.

5.3 Political Participation: Mechanisms. In section 4.3, we showed that the election of a black mayor on a close interracial election causes an increase in the proportion of candidates running for State Deputy born in the mayor's municipality who self-declare as black. This effect was stronger for less experienced candidates (the ones who had never held an elective office), which is a suggestive evidence in favor of a role model effect: indeed, we expect only inexperienced candidates to change their decision of running or not on the election after seeing the success of a black candidate in the municipal election.

Moreover, the fact that we only obtain a significant result for candidates running for State Deputy, and not for Federal Deputy, is also an indirect evidence in favor of this explanation. Generally, the position of Federal Deputy is perceived as higher in the political career – it is also generally more competitive and demands more expensive campaigns, since there are less seats available. Thus, it is less likely that an inexperienced candidate would choose to start his or her political career running for Federal Deputy. We can also see (as shown in Table A.3 on the appendix) that State Deputies are on average younger and more likely to be running for the first time than Federal Deputies. All these information point to the conclusion that, on average, candidates running for Federal Deputy are more experienced than candidates running for State Deputy. For the same reason just discussed, this implies that potential candidates running for State Deputy are more susceptible to having their decision to run or not influenced by a role model. This is consistent with the result we obtained: the election of a black mayor only affects the proportion of black candidates running for State, but not

for Federal Deputy.

Nevertheless, another possible explanation for the results obtained is that the mayor, once elected, favors his or her political group – which might be disproportionately composed of individuals who self-declare as black –, helping these people to also start a political career. To test for this possibility, we run the same analysis as in section 4.3, but excluding candidates running for State Deputy who are from the same party as the elected mayor. The results, shown on Panel B of Table 5, indicate that the effect of the election of a black mayor is even larger for this sub-sample than for the entire population of candidates for State Deputy. Indeed, considering our preferred specification, the election of a black mayor on a close interracial election increases by 13 percentage point the proportion of candidates born in the mayor's municipality running for State Deputy who self-declare as black. This suggests that the results are not explained by the mayor actively helping his or her political allies – as we would expect them to be in the same party as the mayor.

Therefore, there is a considerable amount of evidence indicating that the results obtained on political participation is, at least in part, explained by a role model effect.

5.4 Changes in Self-declaration of Race. The last possible channel that could be (at least partially) explaining the results from section 4 that we will discuss is the possibility that the election of a black candidate as mayor changes some individuals' self-declaration of race. It is possible that the success of a black candidate causes some people to reflect about their own racial identification. Therefore, considering for instance the results related to ENEM enrollment, the increase in black students that we observe could be related to two different movements: on one hand, of students who wouldn't take the exam, but decide to take it after seeing a black candidate being elected; on the other hand, of students who would take the exam anyway, but change their racial self-declaration following the election of a black candidate.

It is unlikely that this "identity" channel explains the entirety of the results for the ENEM enrollment: if it did, we would expect to see a reduction on the number of white students who take ENEM proportional to the increase for black students, which does not happen (see Table 3).

The same can be said about the political participation results: in Table 9, instead of considering the effect of the election of a black candidate on the municipal election on the *proportion* of black candidates running for State Deputy in the following mid-term election, we considered the *number* of candidates running for this position who self-declare as black or white. The RD estimates from this table show that, while the number of black candidates running for State Deputy in the 2018 election increases as a consequence of the election of a black mayor, the number of white candidates is not affected. There is even some weak evidence that the number of white candidates also increases, but in a lower scale.

Therefore, it cannot be that the entire effect obtained is explained by a change in racial identity. It is not possible, however, to rule out this effect of identity change or to decompose our result between changes in behavior (enroll or not on ENEM; be or not a candidate) and changes in identity. Nevertheless, both channels can be interpreted in light of a role model

effect: in one case, the election of a black mayor motivated an individual to reflect about his or her racial declaration (possibly increasing his racial consciousness); in the other case, the election motivated an individual to change his or her behavior directly.

6 Robustness and External Validity.

In this section, we present more evidence on the validity of the RD assumptions in our setting, and discuss the interpretation and external validity of the results.

As pointed out in section 3, the main identification assumption of the RD design is that the expected value of the potential outcomes are continuous around the threshold. In our context, this hypothesis has two main testable implications: first, it must be that municipalities with a close interracial election won by a black candidate (treated) and municipalities with a close interracial election won by a white candidate (control) do not exhibit any discontinuity in pre-treatment variables. This assumption was thoroughly discussed in section 4.1, which shows that we do not have any evidence of discontinuities on pre-treatment variables among treated and control municipalities.

The second testable implication of the RD identification assumption is that there cannot be a perfect sorting around the threshold, i.e., municipalities cannot perfectly manipulate whether they will be treated or control. Theoretically, we do not expect such manipulation to occur: since municipal elections in Brazil are very competitive, it is hard to predict results in advance, specially in close elections. Therefore, no candidate can know for sure if he or she will win or lose the election. Despite this theoretical prediction, we test for manipulation in the data following Cattaneo et al. (2020). In our test, the null hypothesis is that the density of the running variable is continuous at the cutoff (specifically, that the density of candidates who self-declare as black is continuous for candidates that won and lost close elections). Performing the test with our data, we obtain a p-value of 0.8953, i.e., we do not have evidence that it is possible to manipulate the side of the cutoff. This result can be seen graphically on figure 2.

Hence, we have considerable evidence for the validity of the RD identification assumption. Nevertheless, given that RD results may be very sensitive to specification choices, it is important to show that the results are robust to those choices. In this spirit, we report on the appendix all regressions with a full set of covariates. All results remain the same. For our main results in section 4, I also report, in figures 6 and 7, that the results are robust to the choice of bandwidth. The figures show that the point-estimate and confidence intervals are stable across most bandwidths, which is very reassuring.

A final robustness check consisted in re-estimating the model for several placebo cutoffs. Figure A.5 in the appendix shows the distribution of the placebo test statistic when we consider our main outcome, the number of black students who took ENEM in 2017. Specifically, using 100 placebo thresholds for the running variable (all values between -0.5 and 0.5, with an increment of 0.01 each time), we estimate a local linear regression using Calonico et al. (2014)'s method and store the t statistic obtained. Since these regressions consider a placebo

threshold, we would expect to obtain a null treatment effect, differently from what happens when we consider the "true" cutoff. Indeed, figure A.5 shows that the t statistic obtained using the "true" cutoff is among the most extreme ones, which reinforces our result. Note that this placebo test allow us to compute a permutation test p-value, in the spirit of Young (2019). The permutation test p-value implied by our estimates for the effect of an election of a black mayor on the number of black students who took ENEM in 2017 is of 0.079 for a regression without covariates, i.e., the effect is statistically significant at the 10% level. This result is similar to the one obtained using the traditional method, reported on Table 3. For the other outcomes, this method also reinforced the results, so the respective distribution of t-statistics and permutation test p-values are not reported.

All these tests are reassuring and indicate that the RD assumptions and the results obtained are valid in our context. However, it is important to recall that, as in all RD analysis, the parameter that we are estimating represents a local average treatment effect, valid only for municipalities with interracial elections decided by a close margin. This results may not generalize to other municipalities, i.e., it is possible that the effect of the election of a black candidate is different for municipalities that did not have close elections, and even more so for municipalities in which no candidate among the two most voted self-declared as black. Moreover, the municipalities in our sample – the ones that had an interracial election – are not representative of the population of Brazilian municipalities. This can be seen in Appendix Table A.4, which shows that the municipalities in our sample differ from the average Brazilian municipality across several dimensions: they are more likely to be governed by a women, by a younger mayor or by a politician from a left wing party; they also have, on average, a lower GDP and alphabetization rate, as well as are more likely to be on Brazil's North or Northeast regions. Given those differences, it is possible that the results obtained in this paper do not generalize to an average Brazilian municipality. If the effect of the election of a black mayor is stronger, weaker, or equal to the ones obtained is a question for future research.

7 Concluding Remarks and Discussion

This paper presents evidence on what happens after a candidate who self-declares as black wins a close interracial municipal election in Brazil, using data from the 2016 municipal election – the first one in which information on candidate's racial self-declaration was available. We find that the election of a black candidate as mayor causes an increase in the number of high school students who enroll for Brazil's National High School Examination (ENEM). It also causes an increase in the proportion of candidates running for State Deputy born in the mayor's municipality who self-declare as black.

Both results are quantitatively meaningful: for ENEM enrollment, we estimate an increase in enrollment of approximately 30 high school students who self-declare as black – there is virtually no effect for students who self-declare as white; and, for the proportion of black candidates running for State Deputy, the increase is of approximately 15 percentage point, and is stronger when we only consider inexperienced candidates.

The results are at least partially explained by a role model effect: the election of a black candidate as mayor signals to the municipality's population that black individuals may be successful in a political career (more broadly, in a leadership position), which incentivizes investment on education and political participation for individuals who identify with said mayor. We rule out alternative explanations related to a mayor's investment in policy – by showing that there is no evidence that the election of a black mayor increases expenditure on education or culture, nor affects municipal schools' infrastructure –, and related to the mayor providing advantages to his or her political group – by showing that the results are unchanged (or even stronger) when we exclude candidates for State Deputy from the same party as the mayor.

This result is relevant because it shows that having positive examples – or role models – may be an important determinant of behavior and of relevant life choices. The nonexistence of such role models may reinforce inequalities that are already stark in countries such as Brazil. This evidently does not mean that role models are enough to close racial gaps – given that those gaps have deeper historical and sociological roots, and are part of a complex problem related to systemic racism. Nevertheless, the results discussed here show that, on the margin, the existence of role models can be a powerful tool to increase education investment and political participation of groups that are under-represented on these spheres.

More broadly, the results of this paper may be informative to the debate about policies that incentivize the entry of black candidates on politics, such as quotas. This topic has been intensely debated in recent elections in Brazil. The results discussed here illustrate a way in which the election of black candidates may be beneficial to society, which can be understood as an argument in favor of such policies. Nevertheless, the external validity of the RD estimates must be taken into consideration when reaching such a conclusion.

Finally, there is still much to explore in the research of the effect of race on election outcomes, as there are many interesting questions still open. This is specially true in Brazil, given how recent the data of candidates' self-declaration of race is. In particular, it will be interesting to verify if the results reported in this paper hold in the long run and for different election years.

References

Abrucio, Fernando Luiz, "A dinâmica federativa da educação brasileira: diagnóstico e propostas de aperfeiçoamento," *Educação e federalismo no Brasil: combater as desigualdades, garantir a diversidade. Brasília: UNESCO*, 2010, pp. 39–63.

Ajzenman, Nicolás, "The power of example: Corruption spurs corruption," *American Economic Journal: Applied Economics*, 2021, *13* (2), 230–57.

_ , Tiago Cavalcanti, and Daniel Da Mata, "More than Words: Leaders's Speech and Risky Behavior During a Pandemic," 2022.

Alsan, Marcella, Owen Garrick, and Grant Graziani, "Does diversity matter for health? Experimental evidence from Oakland," *American Economic Review*, 2019, *109* (12), 4071–4111.

Arvate, Paulo, Sergio Firpo, and Renan Pieri, "Can women's performance in elections determine the engagement of adolescent girls in politics?," *European Journal of Political Economy*, 2021, p. 102045.

Beaman, Lori, Esther Duflo, Rohini Pande, and Petia Topalova, "Female leadership raises aspirations and educational attainment for girls: A policy experiment in India," *Science*, 2012, *335* (6068), 582–586.

_ , Raghabendra Chattopadhyay, Esther Duflo, Rohini Pande, and Petia Topalova, "Powerful women: does exposure reduce bias?," *The Quarterly journal of economics*, 2009, *124* (4), 1497–1540.

Benjamini, Yoav and Yosef Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal statistical society: series B (Methodological)*, 1995, *57* (1), 289–300.

Brollo, Fernanda and Ugo Troiano, "What happens when a woman wins an election? Evidence from close races in Brazil," *Journal of Development Economics*, 2016, *122*, 28–45.

Broockman, David E, "Black politicians are more intrinsically motivated to advance blacks' interests: A field experiment manipulating political incentives," *American Journal of Political Science*, 2013, *57* (3), 521–536.

Bruce, Raphael, Alexsandros Cavgias, Luis Meloni, and Mário Remígio, "Under pressure: Women's leadership during the COVID-19 crisis," *Journal of development economics*, 2022, *154*, 102761.

Calonico, Sebastian, Matias D Cattaneo, and Rocio Titiunik, "Robust nonparametric confidence intervals for regression-discontinuity designs," *Econometrica*, 2014, *82* (6), 2295–2326.

Campbell, David and Christina Wolbrecht, "See Jane run: Women politicians as role models for adolescents," *The Journal of Politics*, 2006, *68* (2), 233–247.

Campos, Luiz Augusto and Carlos Machado, "A cor dos eleitos: determinantes da sub-representação política dos não brancos no Brasil," *Revista Brasileira de Ciência Política*, 2015, pp. 121–151.

Cattaneo, Matias D, Michael Jansson, and Xinwei Ma, "Simple local polynomial density estimators," *Journal of the American Statistical Association*, 2020, *115* (531), 1449–1455.

_ , Nicolás Idrobo, and Rocio Titiunik, *A practical introduction to regression discontinuity designs: Foundations*, Cambridge University Press, 2019.

Caughey, Devin and Jasjeet S Sekhon, "Elections and the regression discontinuity design: Lessons from close US House races, 1942–2008," *Political Analysis*, 2011, *19* (4), 385–408.

Chong, Alberto and Eliana La Ferrara, "Television and divorce: Evidence from Brazilian novelas," *Journal of the European Economic Association*, 2009, *7* (2-3), 458–468.

Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso, "Patronage and selection in public sector organizations," *American Economic Review*, 2020, *110* (10), 3071–99.

Cook, Lisa D, Janet Gerson, and Jennifer Kuan, "Closing the Innovation Gap in Pink and Black," Technical Report, National Bureau of Economic Research 2021.

Dee, Thomas S, "A teacher like me: Does race, ethnicity, or gender matter?," *American Economic Review*, 2005, *95* (2), 158–165.

Eggers, Andrew, Anthony Fowler, Jens Hainmueller, Andrew Hall, and James Snyder Jr, "On the validity of the regression discontinuity design for estimating electoral effects: New evidence from over 40,000 close races," *American Journal of Political Science*, 2015, *59* (1), 259–274.

Fairlie, Robert W, Florian Hoffmann, and Philip Oreopoulos, "A community college instructor like me: Race and ethnicity interactions in the classroom," *American Economic Review*, 2014, *104* (8), 2567–91.

Fujiwara, Thomas et al., "A regression discontinuity test of strategic voting and Duverger's law," *Quarterly Journal of Political Science*, 2011, *6* (3-4), 197–233.

Genicot, Garance and Debraj Ray, "Aspirations and inequality," *Econometrica*, 2017, *85* (2), 489–519.

_ and _ , "Aspirations and economic behavior," *Annual Review of Economics*, 2020, *12*, 715–746.

Gershenson, Seth, Cassandra Hart, Joshua Hyman, Constance Lindsay, and Nicholas W Papageorge, "The long-run impacts of same-race teachers," *American Economic Journal: Economic Policy*, Forthcoming.

Hartman, Erin, "Equivalence testing for regression discontinuity designs," *Political Analysis*, 2021, *29* (4), 505–521.

Hopkins, Daniel J and Katherine T McCabe, "After it's too late: Estimating the policy impacts of black mayoralties in US Cities," *American Politics Research*, 2012, *40* (4), 665–700.

Hsieh, Chang-Tai, Erik Hurst, Charles I Jones, and Peter J Klenow, "The allocation of talent and us economic growth," *Econometrica*, 2019, *87* (5), 1439–1474.

IBGE, "Desigualdades Sociais por raça ou cor no Brasil," *Estudos e pesquisas. Informação demográfica e sócio-econômica*, 2019, (41).

Imbens, Guido and Karthik Kalyanaraman, "Optimal bandwidth choice for the regression discontinuity estimator," *The Review of economic studies*, 2012, *79* (3), 933–959.

Imbens, Guido W and Thomas Lemieux, "Regression discontinuity designs: A guide to practice," *Journal of econometrics*, 2008, *142* (2), 615–635.

Jensen, Robert and Emily Oster, "The power of TV: Cable television and women's status in India," *The Quarterly Journal of Economics*, 2009, *124* (3), 1057–1094.

Klašnja, Marko and Rocio Titiunik, "The incumbency curse: Weak parties, term limits, and unfulfilled accountability," *American Political Science Review*, 2017, *111* (1), 129–148.

La Ferrara, Eliana, Alberto Chong, and Suzanne Duryea, "Soap operas and fertility: Evidence from Brazil," *American Economic Journal: Applied Economics*, 2012, *4* (4), 1–31.

Lee, David S, "Randomized experiments from non-random selection in US House elections," *Journal of Econometrics*, 2008, *142* (2), 675–697.

Morgenroth, Thekla, Michelle K Ryan, and Kim Peters, "The motivational theory of role modeling: How role models influence role aspirants' goals," *Review of general psychology*, 2015, *19* (4), 465–483.

Nye, John, Ilia Rainer, and Thomas Stratmann, "Do black mayors improve black relative to white employment outcomes? Evidence from large US cities," *The Journal of Law, Economics, & Organization*, 2015, *31* (2), 383–430.

Porter, Catherine and Danila Serra, "Gender differences in the choice of major: The importance of female role models," *American Economic Journal: Applied Economics*, 2019.

Rabelo, Alexandre, Marcos Nakaguma, Fernanda Estevan, and Thiago Lucena, "Non-white mayors and the racial composition of municipal managers," 2022.

Riley, Emma, "Role models in movies: the impact of Queen of Katwe on students' educational attainment," *The Review of Economics and Statistics*, Forthcoming, pp. 1–48.

Rose, Evan K, "A Constructivist Perspective on Empirical Discrimination Research," 2022.

Sen, Maya and Omar Wasow, "Race as a bundle of sticks: Designs that estimate effects of seemingly immutable characteristics," *Annual Review of Political Science*, 2016, *19*, 499–522.

Vogl, Tom S, "Race and the politics of close elections," *Journal of Public Economics*, 2014, *109*, 101–113.

Washington, Ebonya, "How black candidates affect voter turnout," *The Quarterly Journal of Economics*, 2006, *121* (3), 973–998.

Young, Alwyn, "Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results," *The Quarterly Journal of Economics*, 2019, *134* (2), 557–598.

8 Tables

Table 1: Distribution of interracial municipal elections by candidate's self declaration of race, Brazil 2016

Elected \ Runner-up	**White**	***Pardo***	***Preto***	
White	-	770	39	809
Pardo	702	-	-	702
Preto	52	-	-	52
	754	770	39	1563

Notes: The table displays the distribution of results, in terms of candidates' self-declaration of race, of interracial municipal elections in Brazil for the 2016 election.

Table 2: Descriptive Statistics: Sample of interracial municipal elections (2016)

	White Mayor		Black Mayor		
	Mean	Obs	Mean	Obs	p-value
Mayor's characteristics					
Gender (1=Female)	0.1731	809	0.1233	754	0.0056
Age in election day	48.3844	809	47.2493	754	0.0385
Right-wing party	0.2262	809	0.1976	754	0.1668
Left-wing party	0.2151	809	0.2427	754	0.1945
Workers' Party (PT)	0.0482	809	0.0610	754	0.2670
Brazilian Social Democracy Party (PSDB)	0.1298	809	0.1419	754	0.4853
Incumbent	0.2138	809	0.2228	754	0.6684
Elementary School (incomplete)	0.0519	809	0.0517	754	0.9864
Elementary School (complete)	0.0569	809	0.0676	754	0.3791
High School (incomplete)	0.0321	809	0.0252	754	0.4107
High School (complete)	0.2386	809	0.3037	754	0.0038
University (incomplete)	0.0630	809	0.0504	754	0.2796
University (complete)	0.5451	809	0.4960	754	0.0523
Municipality's Characteristics					
Norte Region	0.1162	809	0.1101	754	0.7031
Nordeste Region	0.4858	809	0.5265	754	0.1076
Centro-Oeste Region	0.1112	809	0.1088	754	0.8749
Sul Region	0.0433	809	0.0451	754	0.8605
Sudeste Region	0.2435	809	0.2095	754	0.1088
GDP (2016)	468,625.8109	809	557,329.4098	754	0.4226
Alphabetization rate (2010)	81.3601	809	80.8830	754	0.2673
Population (2010)	23,785.0890	809	26,703.5968	754	0.4962
Proportion of population self-declared as Black (2010)	0.6441	809	0.6530	754	0.2476
Enrollment in High School (white students, 2016)	185.4614	804	184.8107	750	0.9802
Enrollment in High School (black students, 2016)	537.4498	747	627.0914	689	0.4196
Per capita expenditure on Education (2016)	885.6408	773	900.3531	724	0.4185
Per capita expenditure on Culture (2016)	29.1319	676	30.0355	650	0.6717

Notes: The table displays descriptive statistics for several variables at the mayor and/or municipal level. The reported p-value is the p-value of a simple difference of means test between the municipalities where a white and a black candidate were elected, with null hypothesis that the mean of the variable for both groups are equal.

Table 3: The impact of the election of a black mayor on high school students' participation on ENEM, RDD estimates

Dependent Variable:	Black students enrolled in HS who took ENEM in 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−17.37	21.61	34.64**	42.41*	28.94**	39.13*
	(12.88)	(15.29)	(17.3)	(23.37)	(12.45)	(20.66)
Observations	1563	1563	782	427	782	427
Bandwidth	-	-	0.12	0.06	0.12	0.06
Covariates	No	No	No	No	No	No
Average	56.61	56.61	50.36	46.45	50.36	46.45
Dependent Variable:	Black students enrolled in HS who took ENEM in 2018					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−20.35	16.91	29.03*	37.4*	23.94**	35.87*
	(13.41)	(14.66)	(15.16)	(20.59)	(11.25)	(18.79)
Observations	1563	1563	751	405	747	402
Bandwidth	-	-	0.11	0.06	0.11	0.06
Covariates	No	No	No	No	No	No
Average	53.53	53.53	41.17	41.88	41.28	42.14
Dependent Variable:	White students enrolled in HS who took ENEM in 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−10.36*	7.79	8.31	9.07	4.49	7.01
	(5.33)	(8.63)	(8.89)	(12.49)	(6.13)	(8.62)
Observations	1563	1563	828	452	1097	629
Bandwidth	-	-	0.13	0.06	0.18	0.09
Covariates	No	No	No	No	No	No
Average	26.79	26.79	22.29	21.35	21.65	20.35
Dependent Variable:	White students enrolled in HS who took ENEM in 2018					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−10.73*	4.53	4.77	−2.39	2.99	2.13
	(5.55)	(8.95)	(9.43)	(13.67)	(8.27)	(13.94)
Observations	1563	1563	845	462	808	442
Bandwidth	-	-	0.13	0.06	0.12	0.06
Covariates	No	No	No	No	No	No
Average	26.75	26.75	21.43	20.25	21.38	20.5

Notes: The table shows estimates for the effect of the election of a black mayor on educational outcomes. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 4: The impact of the election of a black mayor on political participation, RDD estimates

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.08^{**}	0.00	0.1	0.04	0.08	0.12
	(0.03)	(0.06)	(0.08)	(0.09)	(0.05)	(0.07)
Observations	762	762	375	202	402	219
Bandwidth	-	-	0.12	0.06	0.13	0.06
Covariates	No	No	No	No	No	No
Average	0.58	0.58	0.61	0.56	0.62	0.57
Dependent Variable:	Proportion of candidates running to Federal Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.1^{**}	-0.05	0.04	-0.03	0.08	0.04
	(0.05)	(0.1)	(0.1)	(0.15)	(0.07)	(0.08)
Observations	490	490	308	170	455	346
Bandwidth	-	-	0.16	0.08	0.41	0.2
Covariates	No	No	No	No	No	No
Average	0.53	0.53	0.53	0.6	0.53	0.53

Notes: The table shows estimates for the effect of the election of a black mayor on political participation. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 5: The impact of the election of a black mayor on political participation, RDD estimates, subsamples

Panel A: Excluding candidates that held an elected office since 2008

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.1***	0.06	0.16**	0.09	0.14***	0.16***
	(0.03)	(0.06)	(0.07)	(0.09)	(0.05)	(0.06)
Observations	743	743	338	183	379	207
Bandwidth	-	-	0.11	0.06	0.13	0.06
Covariates	No	No	No	No	No	No
Average	0.59	0.59	0.61	0.55	0.62	0.57

Panel B: Excluding candidates from the same party as elected mayor

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.1***	0.03	0.13*	0.03	0.11**	0.12*
	(0.03)	(0.06)	(0.08)	(0.09)	(0.05)	(0.07)
Observations	754	754	382	209	386	211
Bandwidth	-	-	0.12	0.06	0.13	0.06
Covariates	No	No	No	No	No	No
Average	0.58	0.58	0.61	0.57	0.61	0.57

Notes: The table shows estimates for the effect of the election of a black mayor on political participation for subsamples of the candidates for State Deputy. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 6: The impact of the election of a black mayor on municipal expenditure on education and culture, RDD estimates

Dependent Variable:	*Per capita* municipal expenditure on Education, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	1.51	−41.45	−61.91	−187.92**	−69.36	−149.07**
	(25.12)	(49.23)	(56.29)	(77.31)	(51.05)	(66.3)
Observations	1534	1534	943	527	727	391
Bandwidth	-	-	0.15	0.08	0.11	0.06
Covariates	No	No	No	No	No	No
Average	906.74	906.74	920.3	904.57	921.04	915.61
Dependent Variable:	*Per capita* municipal expenditure on Culture, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−1.44	−4.8	−6.8	−17.91**	−7.97	−17.23**
	(5.42)	(6.54)	(5.33)	(7.01)	(6.57)	(6.81)
Observations	1392	1392	1055	664	688	373
Bandwidth	-	-	0.22	0.11	0.12	0.06
Covariates	No	No	No	No	No	No
Average	33.72	33.72	33.87	33.75	33.99	35.27

Notes: The table shows estimates for the effect of the election of a black mayor on expenditure on education and culture. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 7: The impact of the election of a black mayor on municipal schools' infrastructure, RDD estimates

Dependent Variable:	Proportion of municipal schools with a library or reading room, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	-0.03^{*}	-0.04	-0.05	-0.05	-0.05^{*}	-0.05
	(0.01)	(0.03)	(0.05)	(0.06)	(0.02)	(0.04)
Observations	1563	1563	1200	751	1115	649
Bandwidth	-	-	0.23	0.11	0.19	0.1
Covariates	No	No	No	No	No	No
Average	0.36	0.36	0.34	0.33	0.34	0.33
Dependent Variable:	Proportion of municipal schools with access to the Internet, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	-0.01	0.02	0.00	0.02	0.00	0.00
	(0.02)	(0.05)	(0.06)	(0.07)	(0.04)	(0.06)
Observations	1563	1563	1180	737	1147	696
Bandwidth	-	-	0.22	0.11	0.21	0.1
Covariates	No	No	No	No	No	No
Average	0.58	0.58	0.56	0.55	0.56	0.55

Notes: The table shows estimates for the effect of the election of a black mayor on municipal schools' infrastructure. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 8: The impact of the election of a black mayor on average number of Bolsa Família beneficiaries (households), RDD estimates

Dependent Variable:	Average number of households who are *Bolsa Família* beneficiaries, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−468.38	683.94	1045.77	1535.2*	885.39**	1322.64*
	(423.87)	(429.46)	(672.5)	(875.9)	(403.5)	(687.68)
Observations	1563	1563	899	496	832	453
Bandwidth	-	-	0.14	0.07	0.13	0.06
Covariates	No	No	No	No	No	No
Average	2640.61	2640.61	2523.01	2416.53	2508.79	2486.86

Dependent Variable:	Average number of households who are *Bolsa Família* beneficiaries, 2019					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−433.58	715.84	1110.41	1674.35*	973.42**	1416.59*
	(399.33)	(442.04)	(714.21)	(930.38)	(432.75)	(753.14)
Observations	1563	1563	903	496	787	429
Bandwidth	-	-	0.14	0.07	0.12	0.06
Covariates	No	No	No	No	No	No
Average	2725.38	2725.38	2609.59	2500.93	2581.26	2547.73

Notes: The table shows estimates for the effect of the election of a black mayor on *Bolsa Família* beneficiaries. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table 9: The impact of the election of a black mayor on political participation, RDD estimates

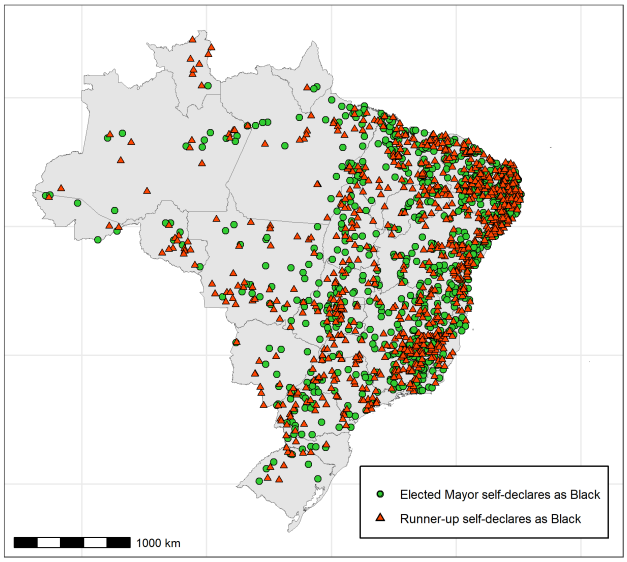
Dependent Variable:	Number of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	-0.96	1.26	1.87^{**}	2.98^{**}	1.61^{**}	2.48^{**}
	(1.05)	(1.08)	(0.86)	(1.35)	(0.67)	(1.18)
Observations	762	762	345	186	369	200
Bandwidth	-	-	0.11	0.06	0.12	0.06
Covariates	No	No	No	No	No	No
Average	2.71	2.71	2.1	1.88	2.25	1.91

Dependent Variable:	Number of candidates running to State Deputy who self-declare as white					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	-0.66	0.52	0.35	1.46^{*}	0.29	0.37
	(0.43)	(0.57)	(0.53)	(0.84)	(0.4)	(0.6)
Observations	762	762	423	239	494	282
Bandwidth	-	-	0.14	0.07	0.17	0.09
Covariates	No	No	No	No	No	No
Average	1.69	1.69	1.33	1.36	1.36	1.32

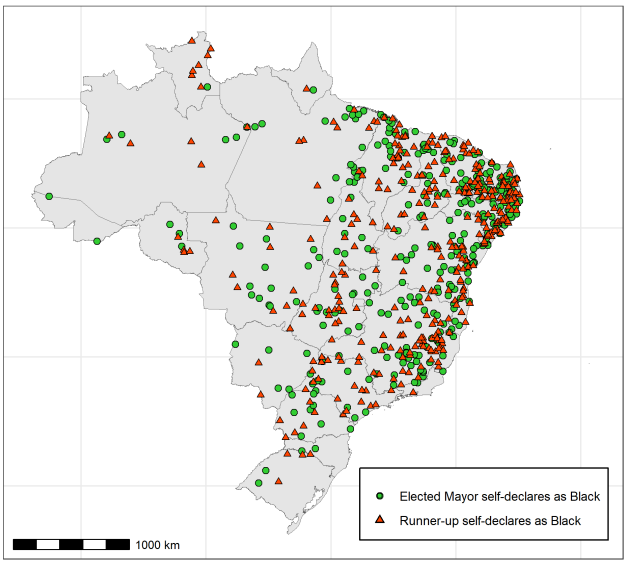
Notes: The table shows estimates for the effect of the election of a black mayor on political participation. I report estimates obtained by OLS of a 1st and of a 4th degree polynomial, as well as local linear regressions. Column (3) uses the optimal bandwidth calculated as in Calonico et al. (2014), using a triangular kernel and reporting bias-corrected estimates and standard errors. Column (4) is the same, but using half the bandwidth. Column (5) uses the optimal bandwidth proposed by Imbens and Lemieux (2008), and column (6) uses half this bandwidth. Standard errors clustered at the state level in parentheses. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

9 Figures

Figure 1: Distribution of municipalities with interracial municipal elections in 2016



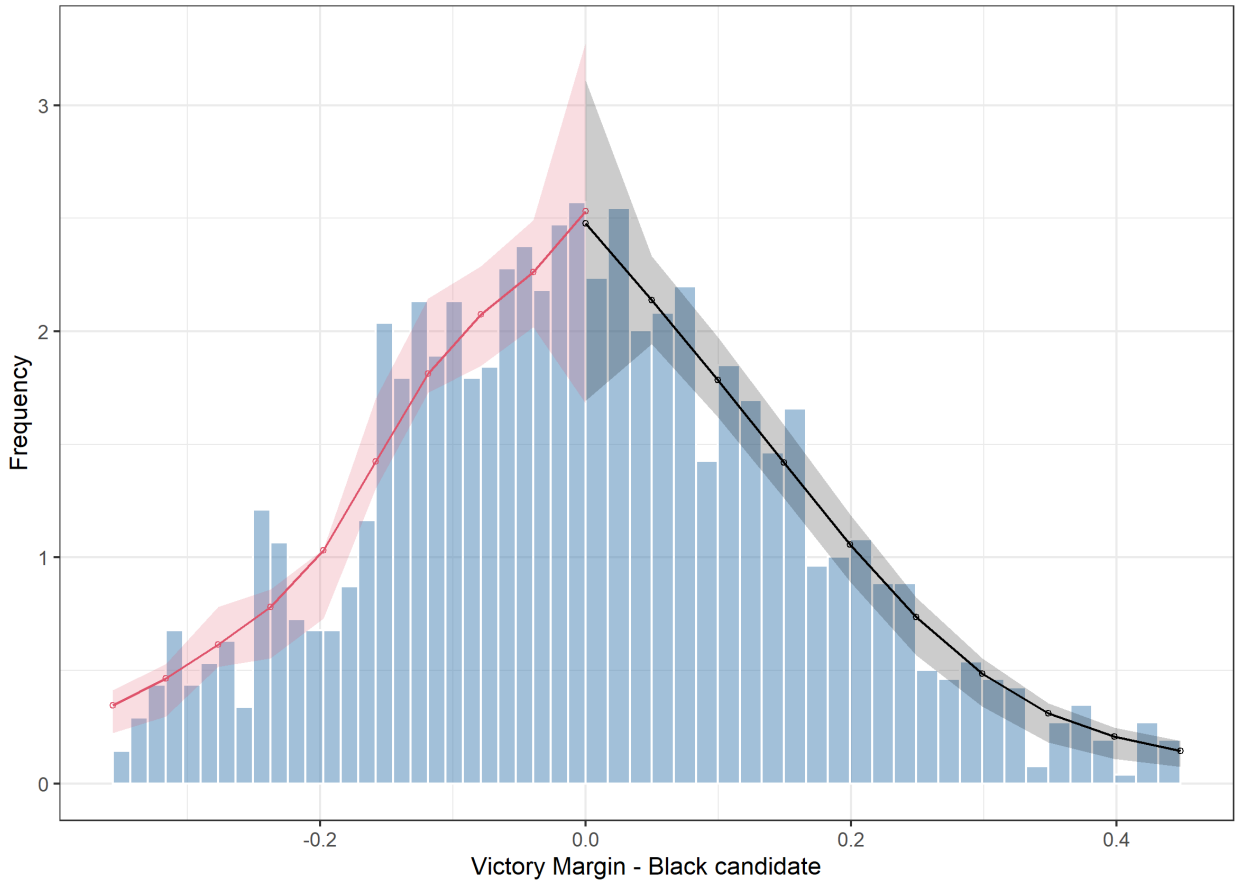
(a) Full sample on interracial elections



(b) Interracial elections decided by a margin of votes non-greater than 0.1 percentage points

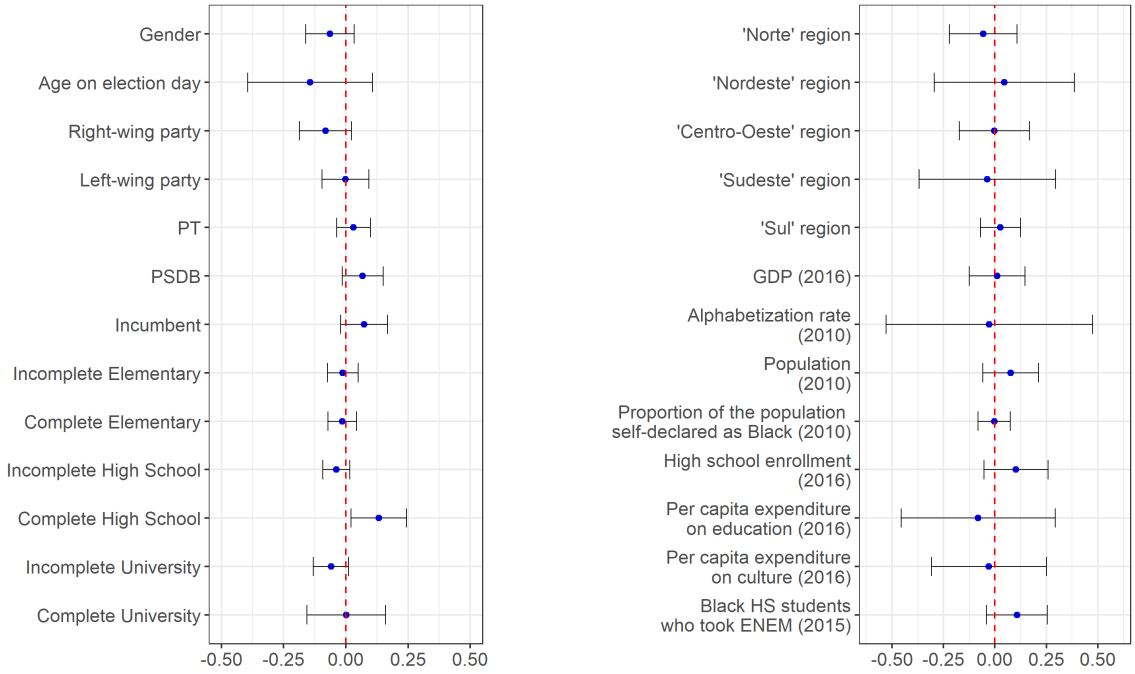
Notes: The figure shows the spatial distribution of interracial election across Brazil's territory. The first figure shows the full sample of interracial elections, while the second shows the municipalities within a 0.1 bandwidth in vote margin.

Figure 2: Manipulation test and Histogram of the vote margin of black candidates



Notes: The figure shows the histogram for the margin of victory of black candidates, defined as the difference in vote share between a black candidate and a white candidate, and a local polynomial density estimate and robust bias corrected confidence intervals computed as described in Cattaneo et al. (2020).

Figure 3: Discontinuities on baseline variables

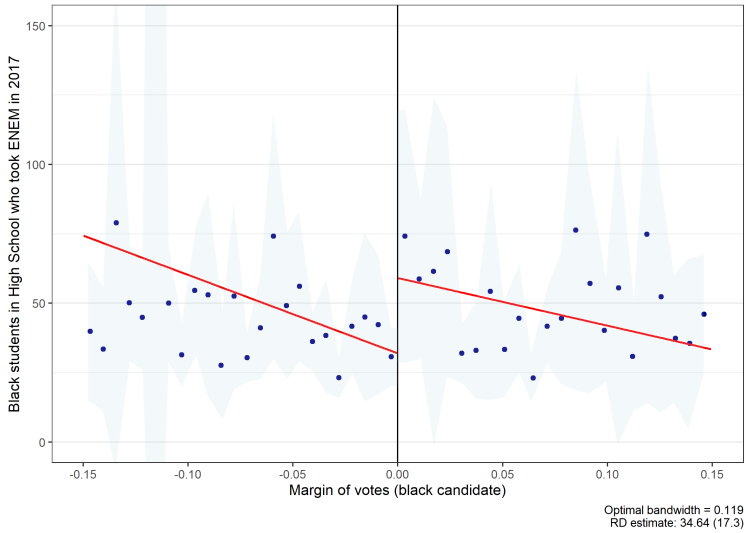


(a) Mayor's baseline characteristics

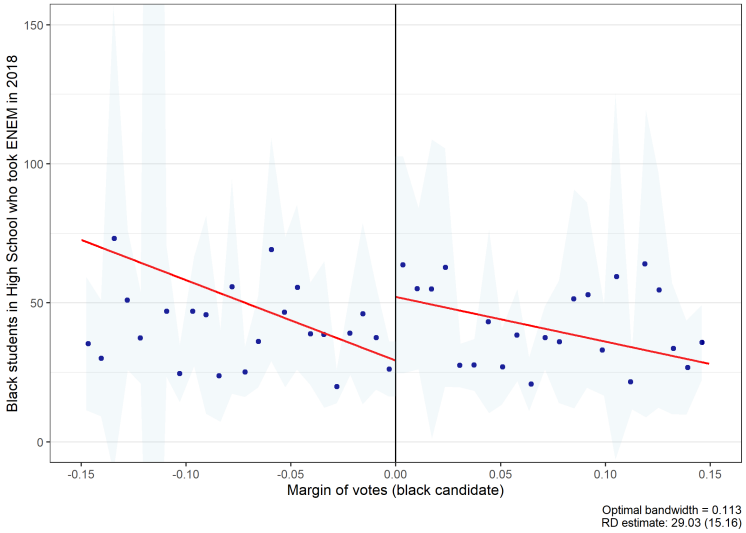
(b) Municipality's baseline characteristics

Notes: The figure shows the estimated discontinuities and 95% confidence intervals on baseline variables between treated and control municipalities, using the local linear regression method from Calonico et al. (2014). Continuous variables were standardized.

Figure 4: The effect of the election of a black mayor on educational outcomes - ENEM participation



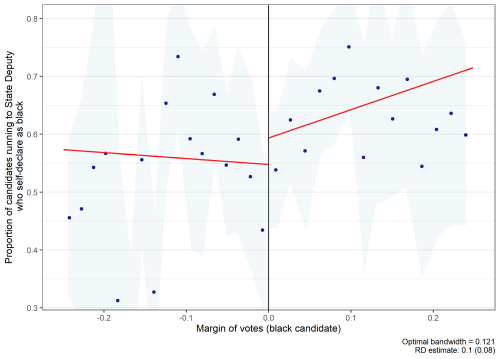
(a) Black high school students who took ENEM in 2017



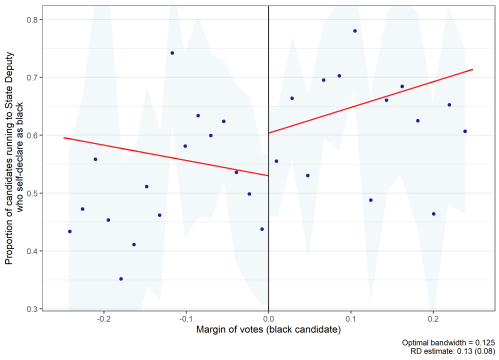
(b) Black high school students who took ENEM in 2018

Notes: The figure shows the mean number of black students in high school who took ENEM in 2017 (panel a) and 2018 (panel b), by bins of vote margin. Municipalities with a positive vote margin are the ones in which a black candidate won the 2016 municipal election. The red lines are first degree polynomials fitted separately at each side of the cutoff (margin of votes of black candidate equal zero). The shaded area represents 90% confidence intervals for each bin.

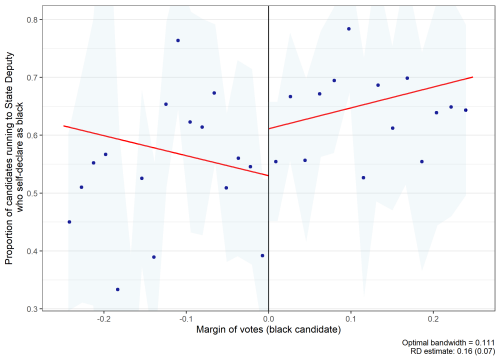
Figure 5: The effect of the election of a black mayor on the proportion of candidates running for state deputy who self-declare as black



(a) Full sample



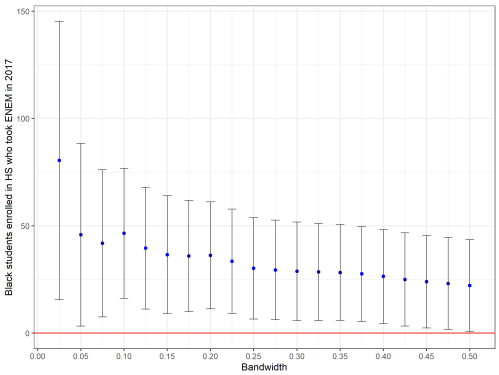
(b) Excluding candidates from the same party as the mayor



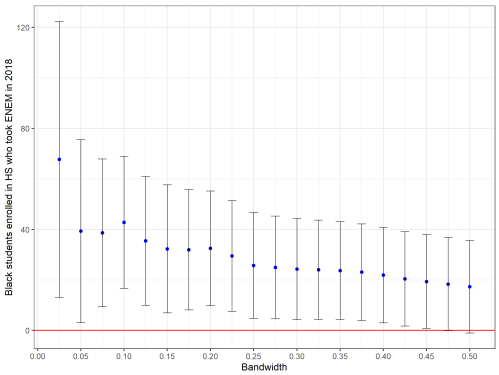
(c) Excluding candidates who have held at least one elective office since 2008

Notes: The figure shows the mean of the proportion of candidates running to State Deputy who self-declare as black by bins of vote margin. Municipalities with a positive vote margin are the ones in which a black candidate won the 2016 municipal election. The red lines are first degree polynomials fitted separately at each side of the cutoff (margin of votes of black candidate equal zero). The shaded area represents 90% confidence intervals for each bin.

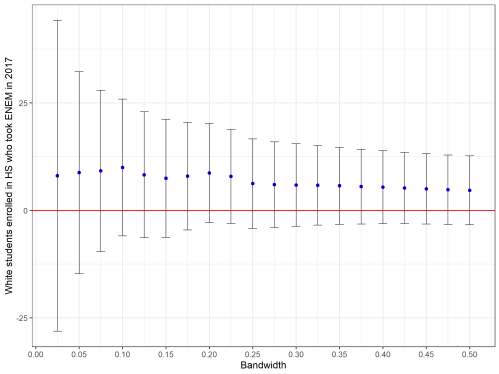
Figure 6: Robustness to the choice of bandwidth: Educational outcomes



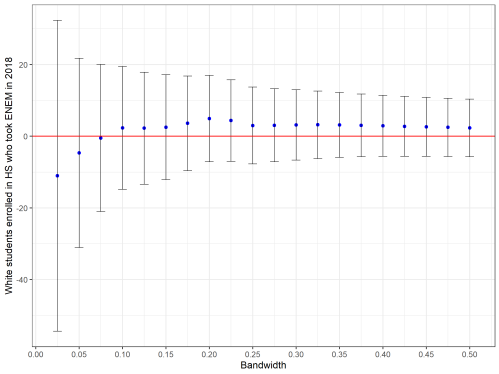
(a) Black students who took ENEM in 2017



(b) Black students who took ENEM in 2018



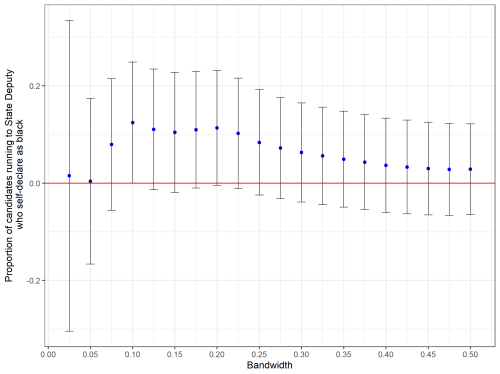
(c) White students who took ENEM in 2017



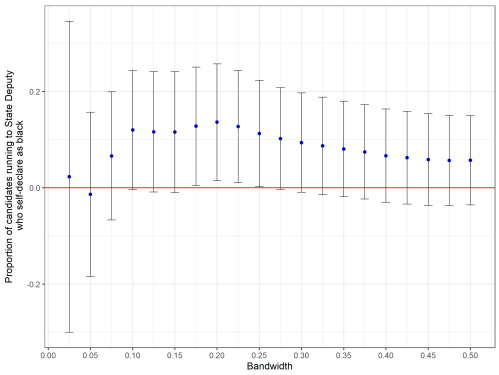
(d) White students who took ENEM in 2018

Notes: The figure shows point-estimates and 90% confidence intervals for the main educational outcomes with different bandwidth choices. The estimates are computed by local linear regression, without covariates, using the bias-corrected method from Calonico et al. (2014). Standard errors are clustered at the state level.

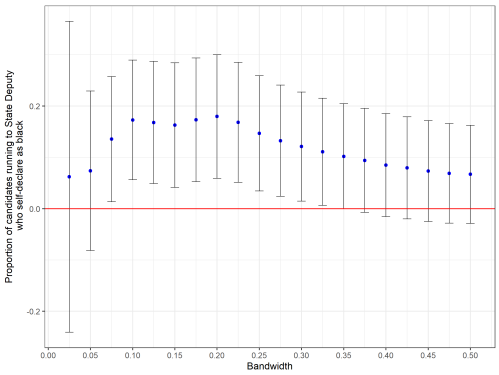
Figure 7: Robustness to the choice of bandwidth: Proportion of candidates running to state deputy in 2018 who self-identify as Black



(a) Full sample



(b) Excluding candidates from the same party as elected mayor



(c) Excluding candidates who have held at least one elective office since 2008

Notes: The figure shows point-estimates and 90% confidence intervals for the main political participation outcomes with different bandwidth choices. The estimates are computed by local linear regression, without covariates, using the bias-corrected method from Calonico et al. (2014). Standard errors are clustered at the state level.

Appendix A. Additional Figures and Tables

Table A.1: Descriptive Statistics: Sample of municipalities with an interracial municipal elections (2016)

	Median	Mean	Std.Deviation	Max	Min	Obs
Mayor's characteristics						
Gender (1=Female)	0.00	0.15	0.36	1.00	0.00	1,563
Age in election day	47.00	47.84	10.88	88.00	21.00	1,563
Right-wing party	0.00	0.21	0.41	1.00	0.00	1,563
Left-wing party	0.00	0.23	0.42	1.00	0.00	1,563
Workers' Party (PT)	0.00	0.05	0.23	1.00	0.00	1,563
Brazilian Social Democracy Party (PSDB)	0.00	0.14	0.34	1.00	0.00	1,563
Incumbent	0.00	0.22	0.41	1.00	0.00	1,563
Elementary School (incomplete)	0.00	0.05	0.22	1.00	0.00	1,563
Elementary School (complete)	0.00	0.06	0.24	1.00	0.00	1,563
High School (incomplete)	0.00	0.03	0.17	1.00	0.00	1,563
High School (complete)	0.00	0.27	0.44	1.00	0.00	1,563
University (incomplete)	0.00	0.06	0.23	1.00	0.00	1,563
University (complete)	1.00	0.52	0.50	1.00	0.00	1,563
Municipality's Characteristics						
Norte Region	0.00	0.11	0.32	1.00	0.00	1,563
Nordeste Region	1.00	0.51	0.50	1.00	0.00	1,563
Centro-Oeste Region	0.00	0.11	0.31	1.00	0.00	1,563
Sul Region	0.00	0.04	0.21	1.00	0.00	1,563
Sudeste Region	0.00	0.23	0.42	1.00	0.00	1,563
GDP (2016)	137,833.00	511,416.93	2,144,549.85	61,168,158.00	13,362.00	1,563
Alphabetization rate (2010)	80.80	81.13	8.49	98.10	59.10	1,563
Estimated Population (2015)	12,025.00	26,961.47	90,050.08	2,921,087.00	1,218.00	1,563
Enrollment in High School (white students, 2016)	61.00	185.15	511.19	8,772.00	1.00	1,554
Enrollment in High School (black students, 2016)	10.00	52.71	570.77	21,189.00	1.00	1,437
Per capita expenditure on Education (2016)	849.41	892.76	348.64	7,191.63	201.17	1,497
Per capita expenditure on Culture (2016)	18.73	29.57	38.67	488.86	0.00	1,326
Outcomes						
Black students enrolled in HS who took ENEM (2017)	22.00	56.70	195.29	5,529.00	0.00	1,563
Black students enrolled in HS who took ENEM (2018)	18.00	53.44	211.03	5,978.00	0.00	1,563
White students enrolled in HS who took ENEM (2017)	9.00	26.38	89.68	1,980.00	0.00	1,563
White students enrolled in HS who took ENEM (2018)	8.00	26.27	94.84	2,132.00	0.00	1,563
Proportion of candidates running to State Deputy who self-declare as black (2018)	0.67	0.59	0.39	1.00	0.00	762
Proportion of candidates running to Federal Deputy who self-declare as black (2018)	0.50	0.53	0.44	1.00	0.00	490
Per capita expenditure on Education (2017)	865.49	908.13	335.95	6,680.94	95.95	1,534
Per capita expenditure on Culture (2017)	23.31	33.43	38.80	484.37	0.01	1,392
Proportion of municipal schools with a library or reading room (2017)	0.30	0.35	0.27	1.00	0.00	1,563
Proportion of municipal schools with access to the internet (2017)	0.57	0.57	0.32	1.00	0.00	1,563
Average number of households who are Bolsa Família beneficiaries (2017)	1,451.50	2,666.75	5,881.96	184,339.17	34.33	1,563
Average number of households who are Bolsa Família beneficiaries (2018)	1,499.50	2,756.19	5,865.42	178,509.17	33.67	1,563

Notes: The table displays descriptive statistics for several variables at the mayor and/or municipal level for the sample of municipalities that had an interracial municipal election in 2016.

Table A.2: Discontinuities in pre-treatment variables, RDD estimates

Panel A: Mayor's characteristics I							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Gender (1=Female)	Age	Right wing	Left wing	PT	PSDB	Incumbent
Black Mayor	−0.063	−1.557	−0.082	−0.002	0.031	0.067	0.073
	(0.05)	(1.39)	(0.05)	(0.05)	(0.03)	(0.04)	(0.05)
90% CI	[−0.145; 0.019]	[−3.845; 0.73]	[−0.169; 0.006]	[-0.081; 0.077]	[−0.026; 0.088]	[−0.002; 0.136]	[−0.006; 0.152]
Obs	1186	1026	965	1098	1034	1075	862
Bandwidth	0.22	0.16	0.15	0.19	0.17	0.18	0.13
Mean	0.15	47.84	0.21	0.23	0.05	0.14	0.22
Panel B: Mayor's characteristics II							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Elementary school (inc.)	Elementary school (comp.)	High School (inc.)	High school (comp.)	University (inc.)	University (comp.)	
Black Mayor	−0.012	−0.015	−0.038	0.132**	−0.059	0.002	
	(0.03)	(0.03)	(0.03)	(0.06)	(0.04)	(0.08)	
90% CI	[−0.063; 0.039]	[−0.063; 0.033]	[−0.084; 0.007]	[0.039; 0.226]	[−0.119; 0.000]	[−0.131; 0.134]	
Obs	1093	894	1040	1008	1135	936	
Bandwidth	0.18	0.14	0.17	0.16	0.2	0.15	
Mean	0.05	0.06	0.03	0.27	0.06	0.52	
Panel C: Municipality's characteristics II							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Norte Region	Nordeste Region	Centro-Oeste Region	Sudeste Region	Sul Region		
Black Mayor	−0.057	0.046	−0.003	−0.037	0.027		
	(0.08)	(0.17)	(0.09)	(0.17)	(0.05)		
90% CI	[−0.195; 0.081]	[−0.24; 0.332]	[−0.146; 0.141]	[−0.315; 0.241]	[−0.055; 0.108]		
Obs	1045	1175	1128	1415	1139		
Bandwidth	0.17	0.22	0.2	0.35	0.2		
Mean	0.11	0.51	0.11	0.23	0.04		
Panel D: Municipality's characteristics II							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	GDP (2016)	Alphabetization Rate (2010)	Estimated Population (2015)	White students enrolled in HS (2016)	Black students enrolled in HS (2016)	Per capita education exp. (2016)	Per capita culture exp. (2016)
Black Mayor	22,210.9	−0.235	6,374.845	31.623	6.902	−28.341	−1.136
	(148,010.4)	(2.18)	(6132.46)	(63.2)	(15.81)	(66.67)	(5.51)
90% CI	[−221,244.5; 265,666.3]	[−3.813; 3.344]	[−3712.152; 16461.842]	[−72.328; 135.574]	[−19.11; 32.914]	[−138.002; 81.32]	[−10.207; 7.934]
Obs	844	1103	842	793	678	998	988
Bandwidth	0.13	0.19	0.13	0.12	0.11	0.17	0.21
Mean	511,416.9	81.13	26,961.5	185.15	52.71	892.76	29.57
Panel E: Pre-treatment Outcomes							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Black HS students who took ENEM in 2015	White HS students who took ENEM in 2015	Proportion of black candidates running for State Deputy in 2014				
Black Mayor	56.822	10.218					
	(40.11)	(15.44)					
90% Confidence Interval	[-9.161; 122.806]	[-15.184; 35.619]					
Obs	858	864					
Bandwidth	0.13	0.13					
Mean	124.91	45.59					

Notes: The table displays RD estimates of the effect of the election of a black candidate on pre-treatment variables. Estimates use the optimal bandwidht calculated as in Calonico et al. (2014). For estimation, I use a triangular kernel and report bias-corrected standard errors (Calonico et al., 2014), clustered at the state level. 90% confidence intervals are also reported. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table A.3: State and Federal Deputies

	Federal Deputies		State Deputies		
	Mean	Std.Dev	Mean	Std.Dev	p-value
Age in election day	48.59	11.67	47.17	11.31	0.00
Gender	0.31	0.46	0.30	0.46	0.32
First time candidate	0.57	0.49	0.60	0.49	0.12
Held an elected office since 2008	0.07	0.25	0.06	0.24	0.28

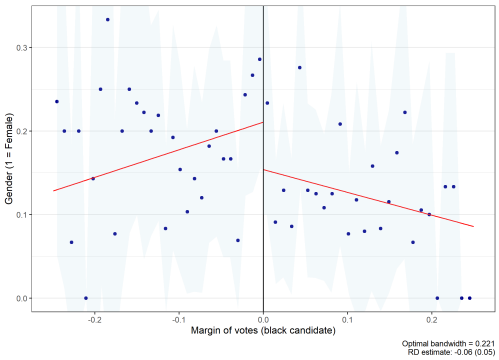
Notes: The table shows a comparison between the characteristics of candidates running for State and Federal Deputy. The reported p-value is the p-value of a test of differences in means between candidates for State and Federal Deputy, for each variable.

Table A.4: Descriptive Statistics: Interracial elections vs Other Elections

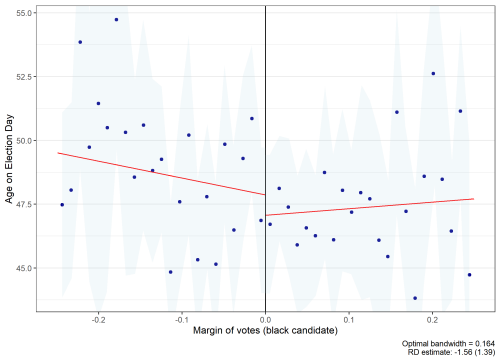
	Other Elections		Interracial Elections		
	Mean	Obs	Mean	Obs	p-value
Mayor's characteristics					
Gender (1=Female)	0.10	3904	0.15	1563	0.00
Age in election day	49.42	3904	47.84	1563	0.00
Right-wing party	0.23	3904	0.21	1563	0.30
Left-wing party	0.18	3904	0.23	1563	0.00
Workers' Party (PT)	0.04	3904	0.05	1563	0.11
Brazilian Social Democracy Party (PSDB)	0.15	3904	0.14	1563	0.30
Incumbent	0.24	3904	0.22	1563	0.15
Elementary School (incomplete)	0.07	3904	0.05	1563	0.03
Elementary School (complete)	0.07	3904	0.06	1563	0.41
High School (incomplete)	0.02	3904	0.03	1563	0.39
High School (complete)	0.26	3904	0.27	1563	0.30
University (incomplete)	0.06	3904	0.06	1563	0.84
University (complete)	0.52	3904	0.52	1563	0.99
Municipality's Characteristics					
Norte Region	0.07	3904	0.11	1563	0.00
Nordeste Region	0.25	3904	0.51	1563	0.00
Centro-Oeste Region	0.07	3904	0.11	1563	0.00
Sudeste Region	0.32	3904	0.23	1563	0.00
Sul Region	0.28	3904	0.04	1563	0.00
GDP (2016)	879742.84	3904	511416.93	1563	0.05
Alphabetization rate (2010)	86.75	3899	81.13	1563	0.00
Estimated Population (2015)	28320.18	3904	26961.47	1563	0.73
Per capita expenditure on Education (2016)	888.11	3804	892.76	1497	0.66
Per capita expenditure on Culture (2016)	29.76	3466	29.57	1326	0.88

Notes: The table displays descriptive statistics for several variables at the mayor and/or municipal level, for municipalities that had or not an interracial election in 2016. The reported p-value is the p-value of a simple difference of means test between the municipalities with an interracial in 2016 or not, with null hypothesis that the mean of the variable for both groups are equal.

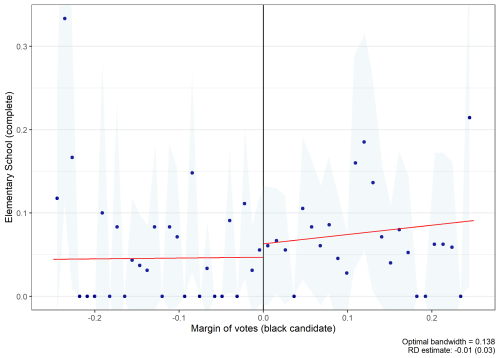
Figure A.1: Discontinuities in mayor's baseline characteristics



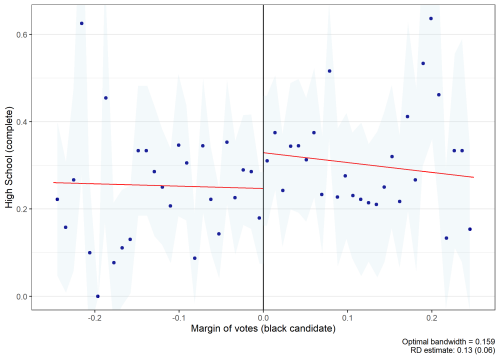
(a) Gender (1=Female)



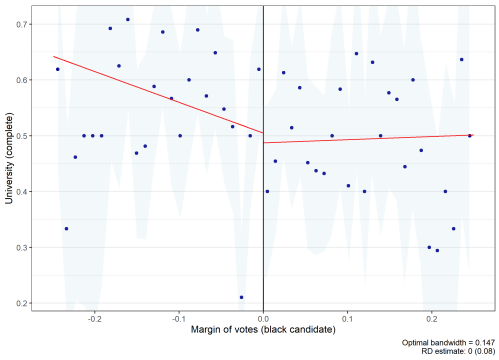
(b) Age in Election Day



(c) Elementary School (complete)



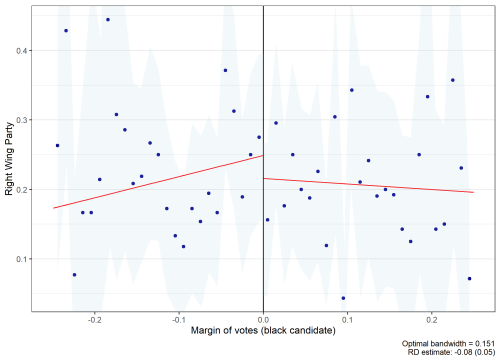
(d) High School (complete)



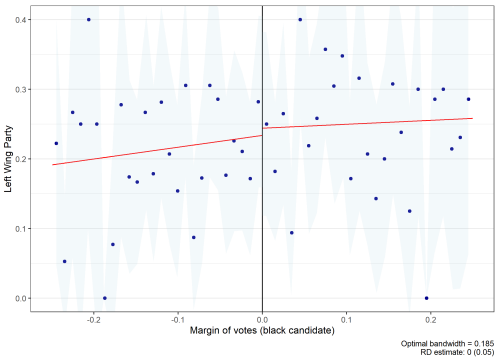
(e) University (complete)

Notes: The figure shows the RD plot for mayor's baseline characteristics. The red lines are first degree polynomials fitted separately at each side of the threshold (margin of votes of black candidate).

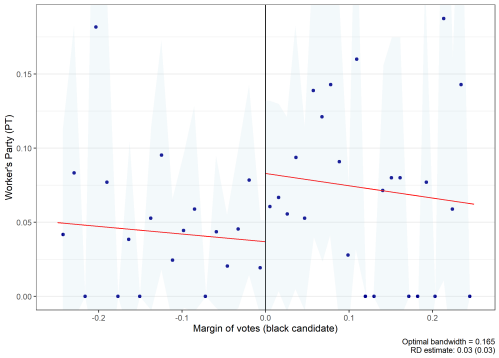
Figure A.2: Discontinuities in mayor's baseline political characteristics



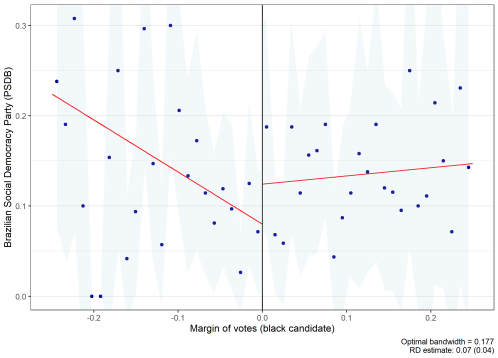
(a) Right wing party



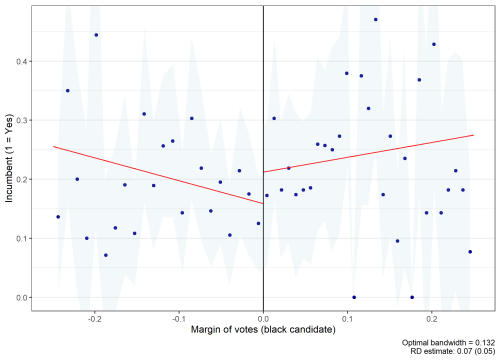
(b) Left wing party



(c) Worker's Party (PT)



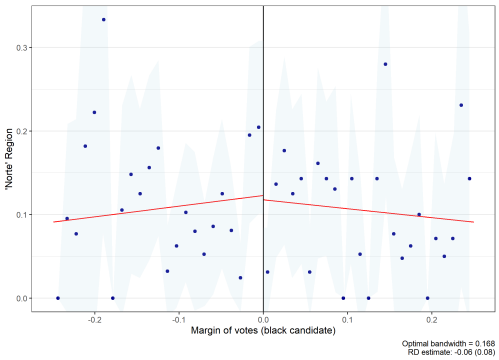
(d) Brazilian Social Democracy Party (PSDB)



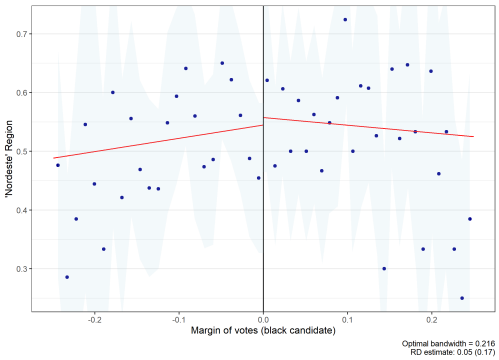
(e) Incumbent (1=Yes)

Notes: The figure shows the RD plot for mayor's baseline characteristics. The red lines are first degree polynomials fitted separately at each side of the threshold (margin of votes of black candidate).

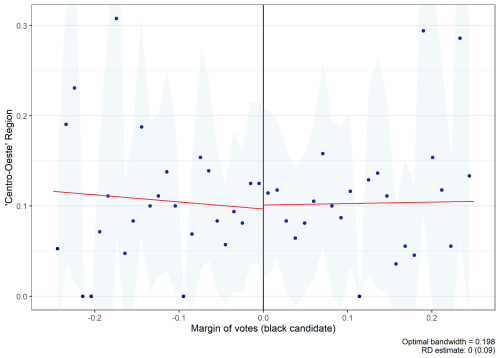
Figure A.3: Discontinuities in municipalities' location (Macro-region)



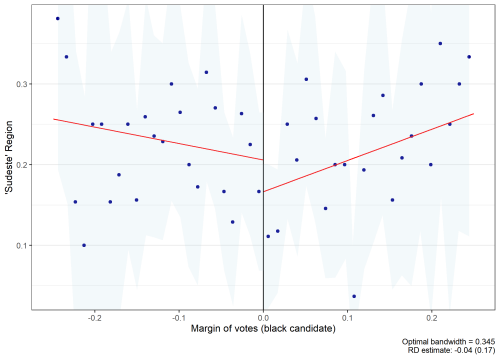
(a) Região Norte



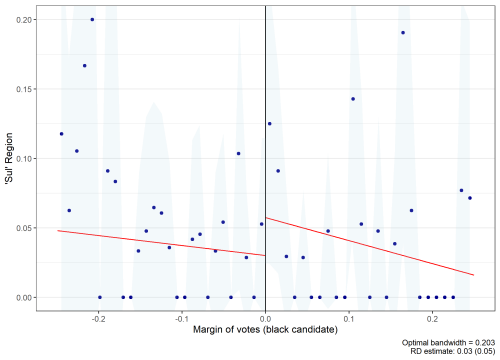
(b) Região Nordeste



(c) Região Centro-Oeste



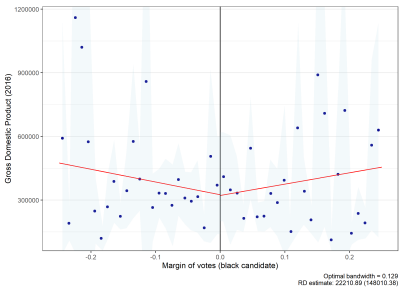
(d) Região Sudeste



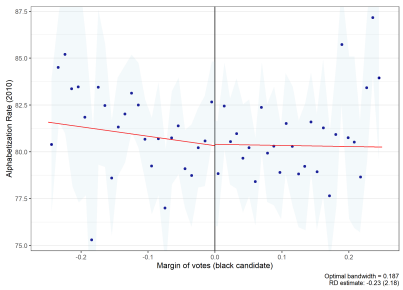
(e) Região Sul

Notes: The figure shows the RD plot for municipality's location (macro-region). The red lines are first degree polynomials fitted separately at each side of the threshold (margin of votes of black candidate).

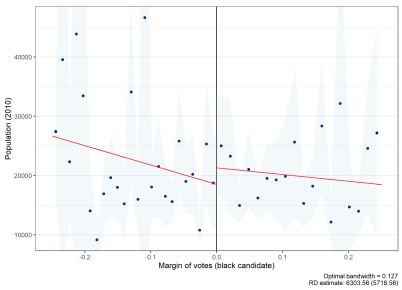
Figure A.4: Discontinuities in municipalities' baseline characteristics



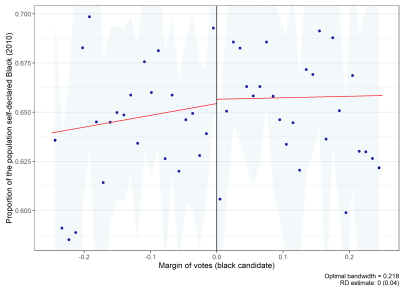
(a) Gross Domestic Product (2016)



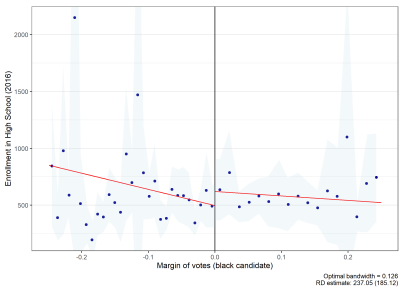
(b) Alphabetization Rate (2010)



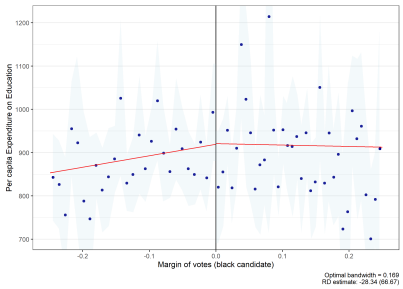
(c) Population (2010)



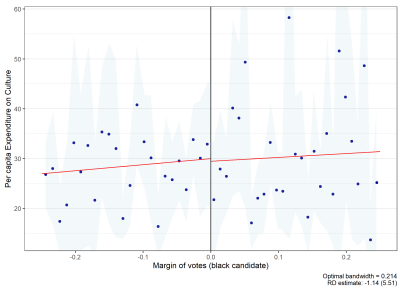
(d) Proportion of the Population self-declared as Black (2010)



(e) Enrollment in High School (2016)



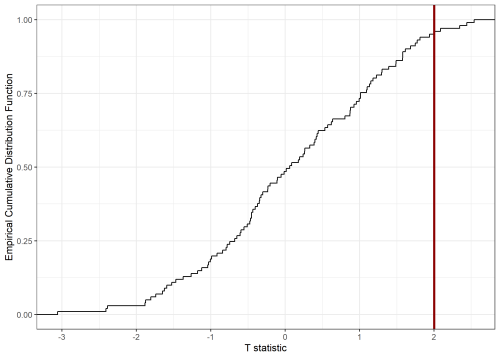
(f) *Per capita* Expenditure on Education (2016)



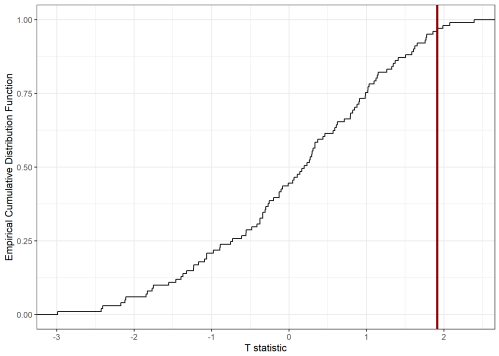
(g) *Per capita* Expenditure on Culture (2016)

Notes: The figure shows the RD plot for municipality's baseline characteristics. The red lines are first degree polynomials fitted separately at each side of the threshold (margin of votes of black candidate).

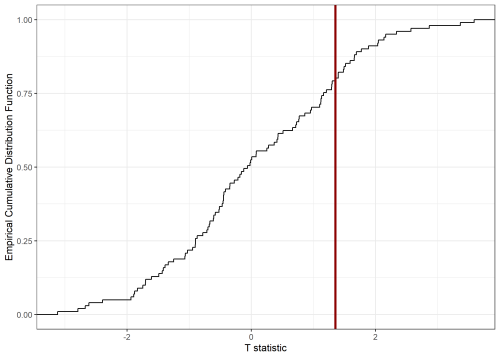
Figure A.5: Placebo test for main outcomes



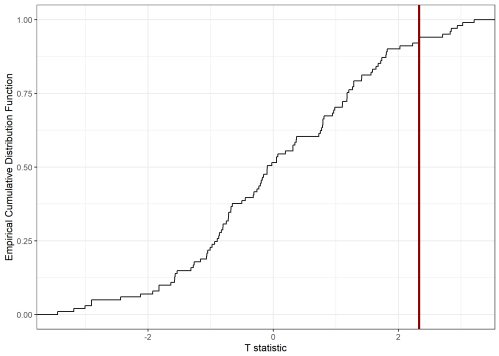
(a) ENEM enrollment by black high school students, 2017



(b) ENEM enrollment by black high school students, 2018



(c) Proportion of candidates running for state deputy who self-declare as black (full sample)



(d) Proportion of candidates running for state deputy who self-declare as black (excluding candidates who have held at least one elective office since 2008)

Notes: The figure shows the empirical cumulative distribution function of the t-statistics of 100 local linear regressions, performed using different placebo thresholds for the running variable. Specifically, for every threshold between -0.5 and 0.5 (with increment of 0.01 between each), we stored the t-statistic of the bias-corrected local linear regression using Calonico et al. (2014)'s method. The red vertical line represents the t-statistic obtained using the "true" cutoff of 0.

Table A.5: The impact of the election of a black mayor on high school students' participation on ENEM, RDD estimates

Dependent Variable:	Black students enrolled in HS who took ENEM in 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−0.69	7.52	13.55*	19.24*	12.31*	21.89*
	(3.77)	(6.73)	(6.95)	(10.7)	(6.42)	(11.33)
Observations	1497	1497	704	372	715	379
Bandwidth	-	-	0.11	0.06	0.11	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	56.61	56.61	46.09	46.25	45.85	46.04
Dependent Variable:	Black students enrolled in HS who took ENEM in 2018					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−2.11	1.52	14.65**	19.44*	9.34**	20.96**
	(4.39)	(7.08)	(6.67)	(10.5)	(4.66)	(9.43)
Observations	1497	1497	659	347	715	379
Bandwidth	-	-	0.1	0.05	0.11	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	53.53	53.53	42.13	41.75	41.17	41.97
Dependent Variable:	White students enrolled in HS who took ENEM in 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−3.53**	2.74	0.8	6.89*	−2.21	0.9
	(1.66)	(4.59)	(3.35)	(3.8)	(2.74)	(4.34)
Observations	1497	1497	769	413	1003	560
Bandwidth	-	-	0.12	0.06	0.17	0.09
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	26.79	26.79	22.38	21.47	22.05	20.34
Dependent Variable:	White students enrolled in HS who took ENEM in 2018					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−3.25*	−1.01	−3.2	−3.35	−4.7	−2.55
	(1.9)	(4.67)	(3.64)	(5.23)	(3.07)	(5.39)
Observations	1497	1497	799	429	863	468
Bandwidth	-	-	0.13	0.06	0.14	0.07
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	26.75	26.75	21.36	20.35	21.64	20.04

Notes: The table shows estimates for the effect of the election of a black mayor on educational outcomes. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects. Estimates without covariates, which are qualitatively similar, are on appendix table 3 $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A.6: The impact of the election of a black mayor on political participation, RDD estimates

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.05	0.02	0.09	0.02	0.11^{*}	0.19^{**}
	(0.04)	(0.07)	(0.08)	(0.1)	(0.06)	(0.08)
Observations	729	729	351	189	375	200
Bandwidth	-	-	0.12	0.06	0.13	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.58	0.58	0.61	0.56	0.61	0.57
Dependent Variable:	Proportion of candidates running to Federal Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.09	-0.02	0.06	-0.15	0.06	0.1
	(0.06)	(0.11)	(0.1)	(0.15)	(0.09)	(0.13)
Observations	472	472	232	124	357	220
Bandwidth	-	-	0.12	0.06	0.23	0.12
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.53	0.53	0.55	0.6	0.53	0.56

Notes: The table shows estimates for the effect of the election of a black mayor on political participation. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariates are on appendix table 4. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A.7: The impact of the election of a black mayor on political participation, RDD estimates, subsamples

Panel A: Excluding candidates that held an elected office since 2008

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.07*	0.09	0.15*	0.05	0.18***	0.23***
	(0.04)	(0.07)	(0.08)	(0.09)	(0.06)	(0.07)
Observations	710	710	335	175	361	194
Bandwidth	-	-	0.11	0.06	0.13	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.59	0.59	0.62	0.55	0.62	0.57

Panel B: Excluding candidates from the same party as elected mayor

Dependent Variable:	Proportion of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	0.07*	0.04	0.11	0.03	0.13**	0.19**
	(0.04)	(0.07)	(0.08)	(0.1)	(0.06)	(0.08)
Observations	721	721	360	191	365	197
Bandwidth	-	-	0.12	0.06	0.13	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.58	0.58	0.61	0.56	0.61	0.57

Notes: The table shows estimates for the effect of the election of a black mayor on political participation for subsamples of the candidates for State Deputy. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariates are on appendix table 5. $^{*}p < 0.1;^{**}p < 0.05,^{***}p < 0.01$.

Table A.8: The impact of the election of a black mayor on municipal expenditure on education and culture, RDD estimates

Dependent Variable:	*Per capita* municipal expenditure on Education, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−7.27	−1.86	−15.91	−41.09	−17.08	−42.38
	(7.76)	(27.61)	(21.19)	(30.81)	(26.33)	(43.26)
Observations	1482	1482	918	508	830	449
Bandwidth	-	-	0.15	0.08	0.14	0.07
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	906.74	906.74	919.49	903.64	918.98	911.37
Dependent Variable:	*Per capita* municipal expenditure on Culture, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−2.17	−3.89	−5.02	−15.43*	−6.55	−11.57
	(4.96)	(6.99)	(6.53)	(8.04)	(6.93)	(7.3)
Observations	1347	1347	814	456	746	412
Bandwidth	-	-	0.15	0.07	0.13	0.07
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	33.72	33.72	33.31	34.41	33.74	34.9

Notes: The table shows estimates for the effect of the election of a black mayor on expenditure on education and culture. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariate are on appendix table 6. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A.9: The impact of the election of a black mayor on municipal schools' infrastructure, RDD estimates

Dependent Variable:	Proportion of municipal schools with a library or reading room, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−0.03	−0.06*	−0.05	−0.02	−0.05	−0.04
	(0.02)	(0.03)	(0.04)	(0.05)	(0.03)	(0.05)
Observations	1497	1497	1185	744	922	512
Bandwidth	-	-	0.24	0.12	0.15	0.08
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.36	0.36	0.34	0.33	0.34	0.33
Dependent Variable:	Proportion of municipal schools with access to the Internet, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−0.01	0	0.01	0.06	0.00	0.02
	(0.02)	(0.04)	(0.05)	(0.05)	(0.04)	(0.06)
Observations	1497	1497	1191	749	891	491
Bandwidth	-	-	0.24	0.12	0.15	0.07
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	0.58	0.58	0.57	0.56	0.56	0.56

Notes: The table shows estimates for the effect of the election of a black mayor on municipal school's infrastructure. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariates are on appendix table 7. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A.10: The impact of the election of a black mayor on average number of Bolsa Família beneficiaries (households), RDD estimates

Dependent Variable:	Average number of households who are *Bolsa Família* beneficiaries, 2017					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	69.18	346.49**	301.47	75.21	229.51**	240.94*
	(90.36)	(127.5)	(202.14)	(295.93)	(110.98)	(132.36)
Observations	1497	1497	666	352	1096	664
Bandwidth	-	-	0.1	0.05	0.21	0.1
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	2640.61	2640.61	2457.36	2451.87	2442.72	2456.11

Dependent Variable:	Average number of households who are *Bolsa Família* beneficiaries, 2019					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	102.24	390.43**	363.11	210.44	219.07*	303.19*
	(95.41)	(146.99)	(260.96)	(379.17)	(121.8)	(162.01)
Observations	1497	1497	691	362	1066	621
Bandwidth	-	-	0.11	0.05	0.19	0.1
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	2725.38	2725.38	2539.63	2517.34	2546.82	2542.09

Notes: The table shows estimates for the effect of the election of a black mayor on *Bolsa Família* beneficiaries. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariates are on appendix table 8. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.

Table A.11: The impact of the election of a black mayor on political participation, RDD estimates

Dependent Variable:	Number of candidates running to State Deputy who self-declare as black					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−0.54	1.83	1.09*	1.38*	1.13**	1.84***
	(0.5)	(1.29)	(0.56)	(0.81)	(0.45)	(0.7)
Observations	729	729	314	165	380	204
Bandwidth	-	-	0.1	0.05	0.13	0.06
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	2.71	2.71	2.13	1.9	2.19	2.08
Dependent Variable:	Number of candidates running to State Deputy who self-declare as white					
Specification	OLS	Polynomial (4th degree)	LLR1 (CCT)	LLR2 (CCT)	LLR3 (IK)	LLR4 (IK)
Black Mayor	−0.27	0.64	0.04	0.81	0.07	0.01
	(0.27)	(0.53)	(0.39)	(0.58)	(0.26)	(0.38)
Observations	729	729	386	209	471	270
Bandwidth	-	-	0.13	0.07	0.17	0.09
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Average	1.69	1.69	1.31	1.34	1.36	1.33

Notes: The table shows estimates for the effect of the election of a black mayor on political participation. Standard errors clustered at the state level in parentheses. All regressions control for municipal GDP in 2016, estimated population, alphabetization rate, number of students enrolled in high school in 2016 and *per capita* expenditure on education, and include state fixed effects as well. Estimates without covariates are on appendix table 4. $^{*}p < 0.1$;$^{**}p < 0.05$,$^{***}p < 0.01$.